

WIRTHSIM

VERSION V1.7.0



WIRTHSIM Professional
embedded simulation

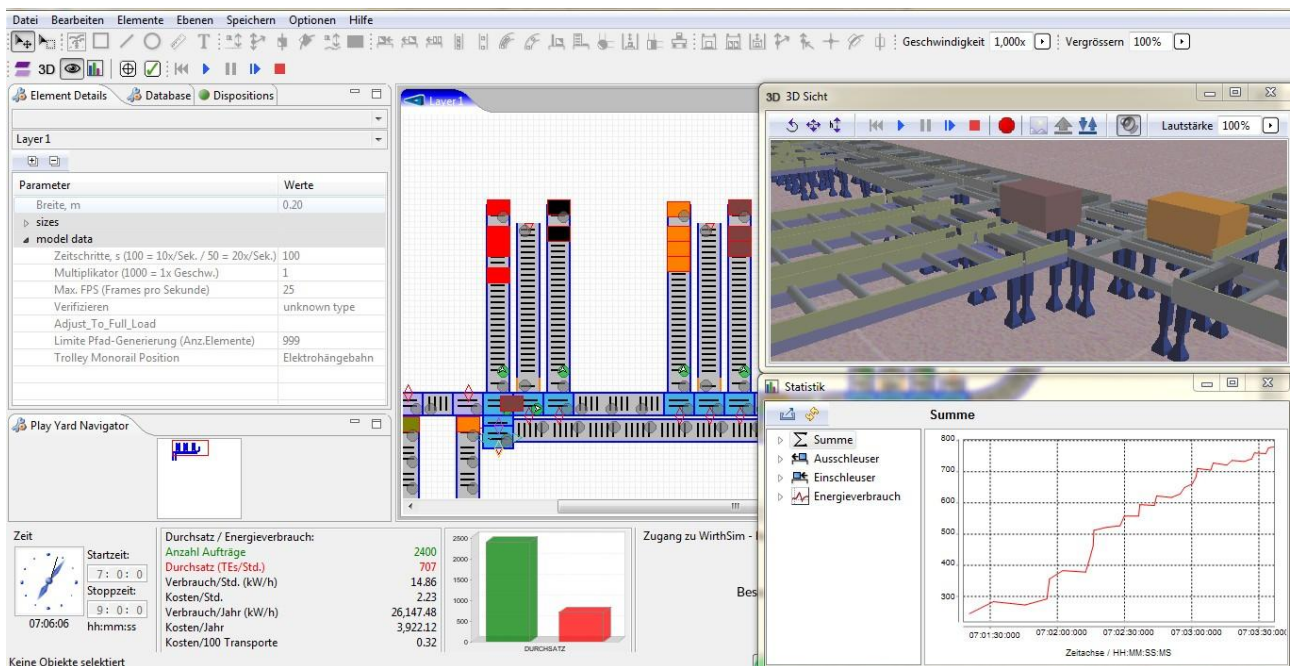


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1 WirthSim Overview

WirthSim is a unique freeware program for simulating automated storage and transport systems. It allows anyone to create complex material flow simulations without any prior software knowledge. WirthSim enables the rapid creation of a prototype during the sales phase as well as detailed modeling during the engineering process. Elements such as conveyors can be selected from the menu and then placed in the layout. As soon as two connection points from two different elements are placed on top of each other in the layout, they connect automatically. No definition is necessary.

WirthSim also offers the following features as a world first:

Eco-Sim

WirthSim supports the definition of individual power consumption and the entry of costs per kW/h for every element in the application. Thus, WirthSim not only displays maximum throughput figures but also estimates annual energy costs. Eco-Sim is rounded out by the identification of cost-intensive peak power consumption.

Audio-Sim

The material handling elements and storage and retrieval machines feature realistic sounds from a live production environment. This means you can not only view the layout in 3D, but also hear it! Experience, for example, storage and retrieval machines in surround sound quality.

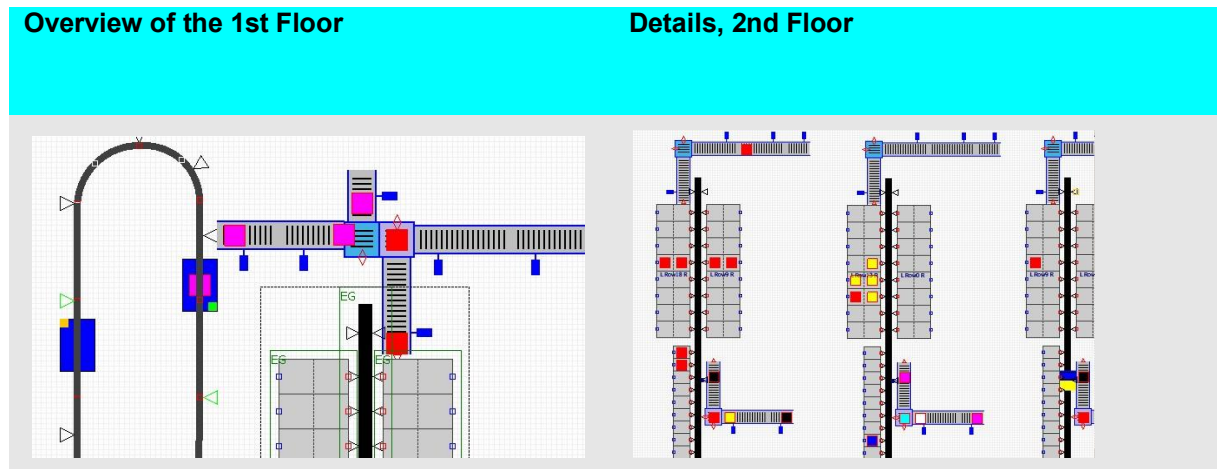
1.1 WirthSim Overview

Supported Operating Systems	Windows 10 and 11
General	2D Design, 3D View, and Statistics
Language Support	German and English
Operation	Start, Stop, and Step Operations
Time-lapse	Up to 100 times faster than in real life *1
Elements	Unlimited number of elements *1
Stop adding elements	Transport units can be stopped (simulation of error conditions)
Floors	n floors / levels

Videos	Recording Movies in 3D View
Times / Dependencies	Jobs are based on time definitions (00:00 - 23:59)
MS Excel Interfaces	Definitions and jobs can be exported to and imported from MS Excel.
2D Image Files	2D image files can be imported for transport units (textures in 3D view) and static elements in the following 2D formats: (*.jpg, *.gif, *.png, *.bmp, or *.ico).
3D image files	Formats *.3ds and *.ac (AC3D) can be imported as transport units or static elements (3D view).
Eco-Sim	Recording and display of energy consumption & peak values (peak current)
Audio-Sim	Audiovisual visualization in 3D
Print	Statistics and Layouts
Installation	WirthSim is automatically installed with all relevant technologies (Java / Java3D).

*1 Depends on computer performance. **WirthSim is a high-performance application** that requires suitable computer hardware.

3 Creating Layouts in 2D



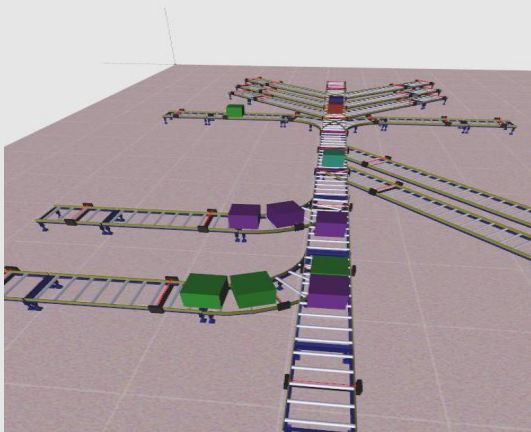
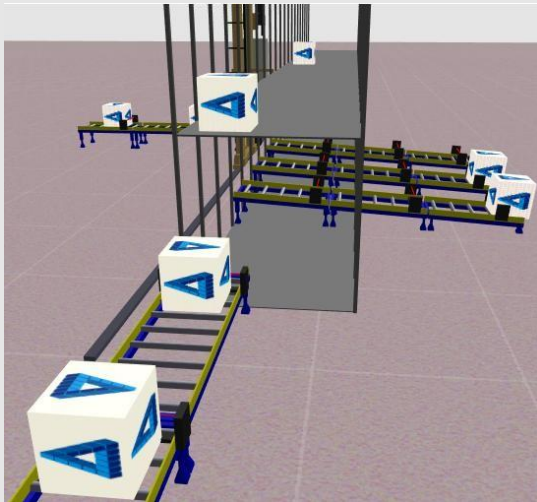
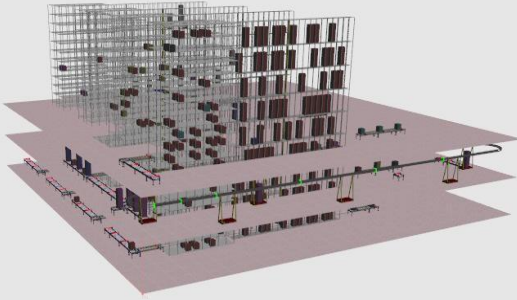
The layout of the WirthSim application is created in a 2D environment. Elements such as conveyors, etc., can be selected from a menu bar and then placed in the layout. Each element contains connection points for linking it to others. As soon as a connection point is placed near another, they are automatically connected.

Using an “elevation meter,” specific elevation ranges can be shown or hidden. This allows for the creation of conveyor routes that run one above the other.

4 3D view of the layouts

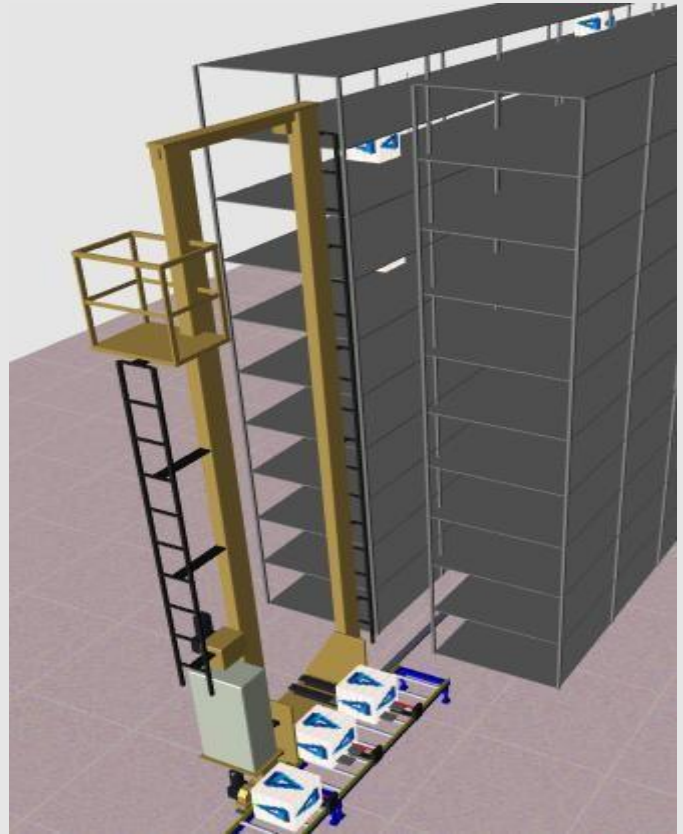
3D View

Layout across multiple levels



3D view

RBG and shuttle systems

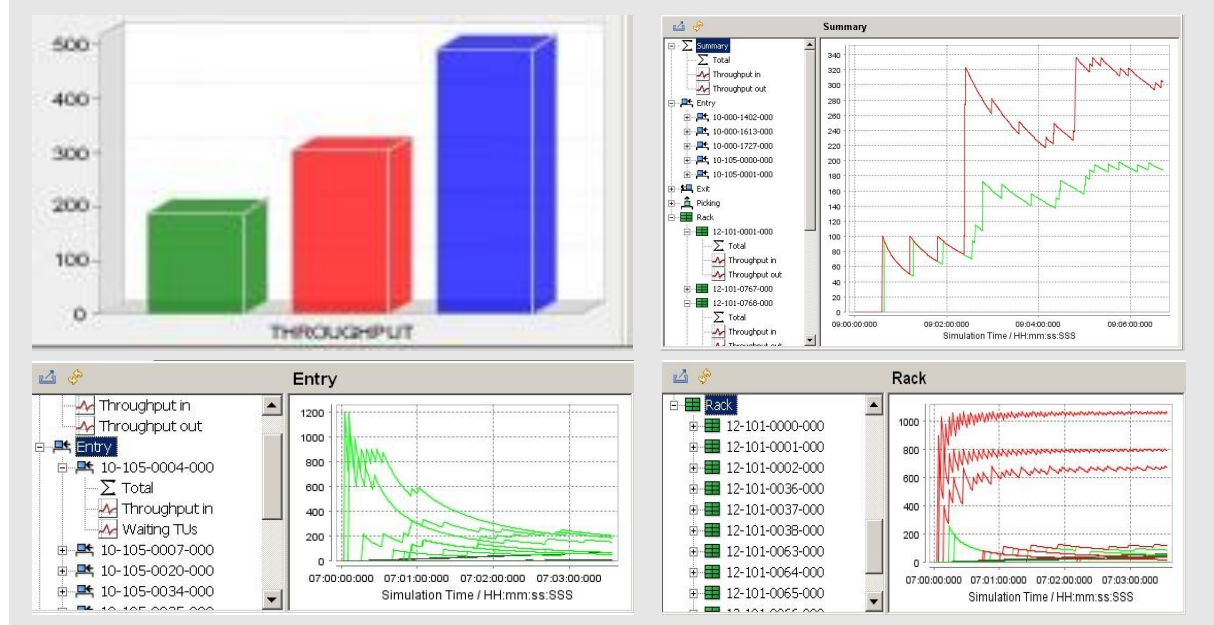


In addition to the 2D development view, it is possible to view the layout in 3D. The user can virtually walk through the simulation model, gaining a realistic impression of the system. Any viewing angle is possible.

5 Statistics

Overview

Details



The Statistics dialog box allows users to centrally verify system performance at any point. Additionally, users can export the data to a Microsoft Excel file or as an image.

Please note that the statistics feature must first be enabled after installation. This feature slows down the application and should only be enabled when evaluating throughput.

7 Using the Application

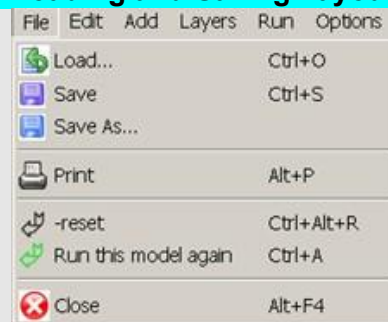
This overview will help you get started with the WirthSim application.

Layout data is saved to the hard drive as *.xml or *.bin files and loaded from there.

Furthermore, it is possible to export and save transport, database (Oracle and MySQL), and statistical data from WirthSim in MS Excel format (*.xls). The following steps describe how to load a simulation layout and then start it in WirthSim.

I. The layout is loaded from the hard disk (*.xml or *.bin format)

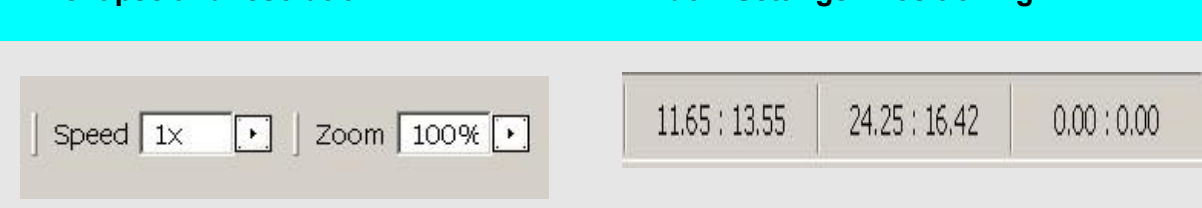
Loading and Saving Layout Data



II. Adjusting the time and layout resolution (optional)

Time-lapse and resolution

Window Settings / Positioning...



Time resolution: '1x' = real time. '5x' = five times faster than in reality (only if the computer hardware allows it).

Resolution: '100%' displays a layout area (window) of approximately 25 m × 16 m. The dimensions can be adjusted at any time. The current size of the window, the window's position, and the cursor position are displayed in the lower-right corner of the application.

III. Adjusting the Start and Stop Times (Optional)

Start and Stop Times



The start and stop settings are used to simulate up to 24 hours. All transport orders used or defined refer to this time definition.

IV. Start, Pause, Stop... Usage

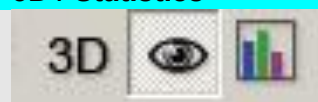
Model Control



Start, Pause, Step, Stop, and Replay the simulation model just like a CD player. The three icons (all three feature the same logo) on the right side are used to save and load “history” data. The WirthSim application supports the use of “random number generators,” e.g., for sorting (profile checking) and storing transport units (racks). After a simulation run is complete, this functionality allows you to exclude these “random events” by loading the “history” file or to run the simulation exactly the same way again.

V. 3D view of the layout and use of the statistics dialog

3D / Statistics



3D: Clicking the 3D button opens the 3D view of the layout.

Eye: Clicking the “Eye” button disables the 2D display of the model. This allows the computer’s CPU to use its processing power exclusively (without graphics) for calculating the model. This feature is often used when employing the time-lapse function.

Statistics: Clicking the Statistics button opens a new window. This window contains an

overview as well as all details regarding system throughput. Statistics data can be saved as an MS Excel file and/or an image file.

8 Creating a Layout

This chapter describes the process for creating a simulation layout in the WirthSim™ application in eight simple steps.

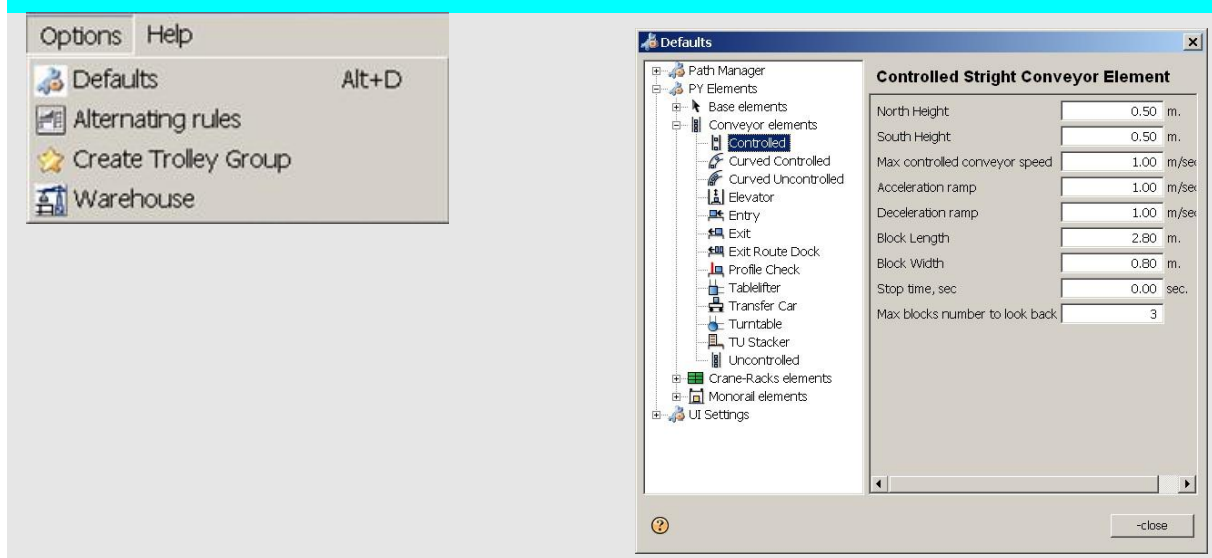
I. SCIR Model

The simulation of throughput is generally based on sound planning and, building on that, on a conveyor and storage layout.

II. Defining the Basic Settings (Element and Layout Data)

Options/Defaults

Defining the default settings for all elements and the layout, such as length, height, width, speed, etc.



Definition of speeds, accelerations, deceleration ramps, lengths, widths, heights... and the layout data such as the length, width, and height(s) of the individual floors

=> After the definition, but before placing or drawing the first element, the layout is saved as a “template.” It can be easily loaded at a later time, and the definition corresponds, for example, to a specific supplier, etc.

III. Adjusting the start and stop times

Adjusting the start and stop times

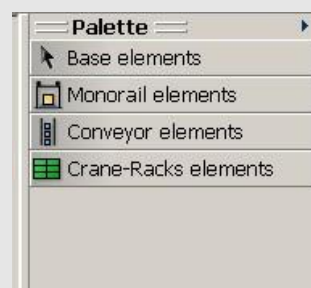


IV. Create Layout

Element list at the top



List of elements on the right



Elements such as conveyors, etc., can be selected from the element list at the top or from the

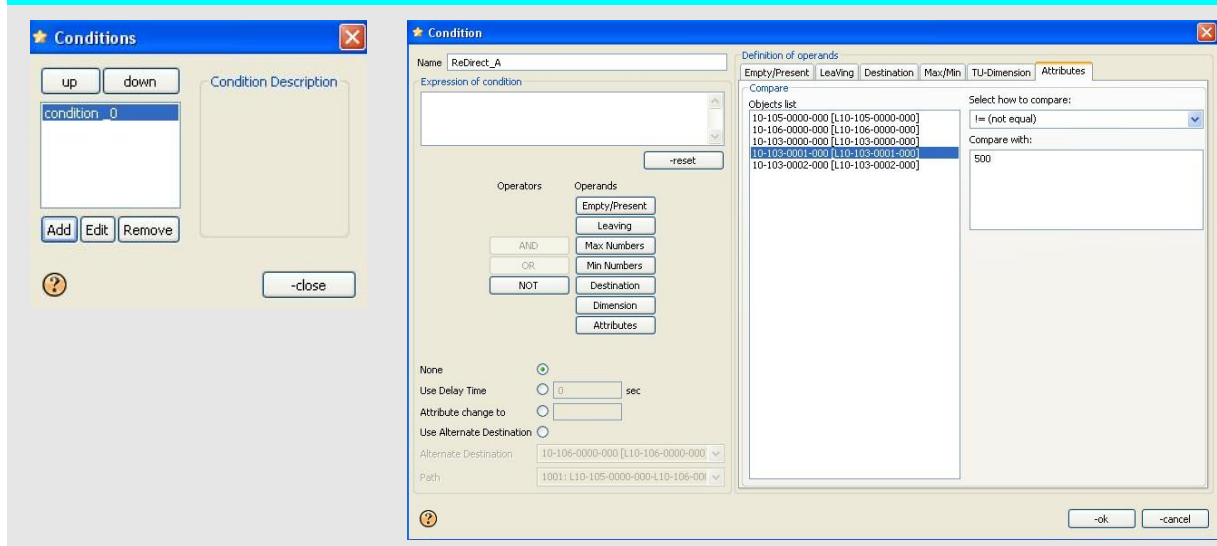
“palette” on the right to then place them in the layout.

Please note:

Pay attention to the conveyor directions, which are indicated by a triangle at the start/end of each conveyor. Otherwise, the application cannot "path" between a source (feeder/rack) and a destination (also called a sink).

V. Definition of Rules/Conditions (Optional)

Definition of Rules and Conditions



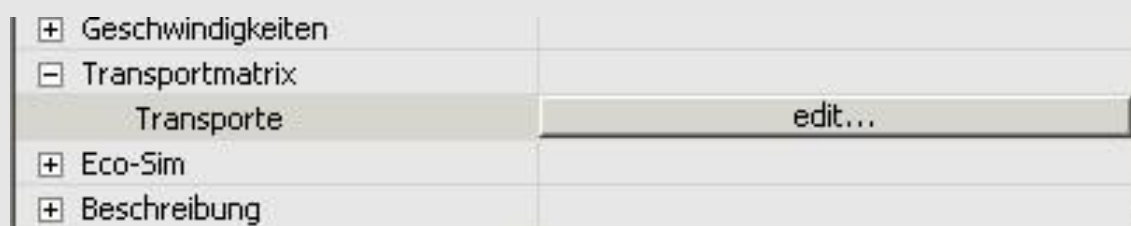
Allows you to define complex dynamic dependencies in the layout without having to write a single line of code. These dependencies can also be adjusted while the simulation is running.

VI. Definition of Transport Orders

Transport orders can be defined for feeders and racks.

Defining Transport Orders at Feeders

For feeders, the dialog opens when you click the “edit...” button on the left in the "Transports" section

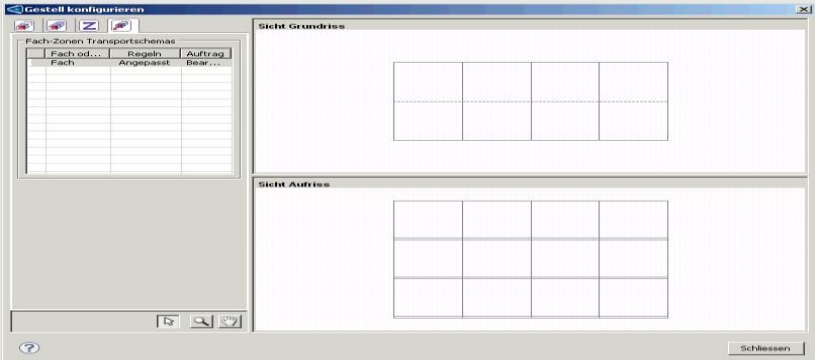


Defining Transport Orders for Racks

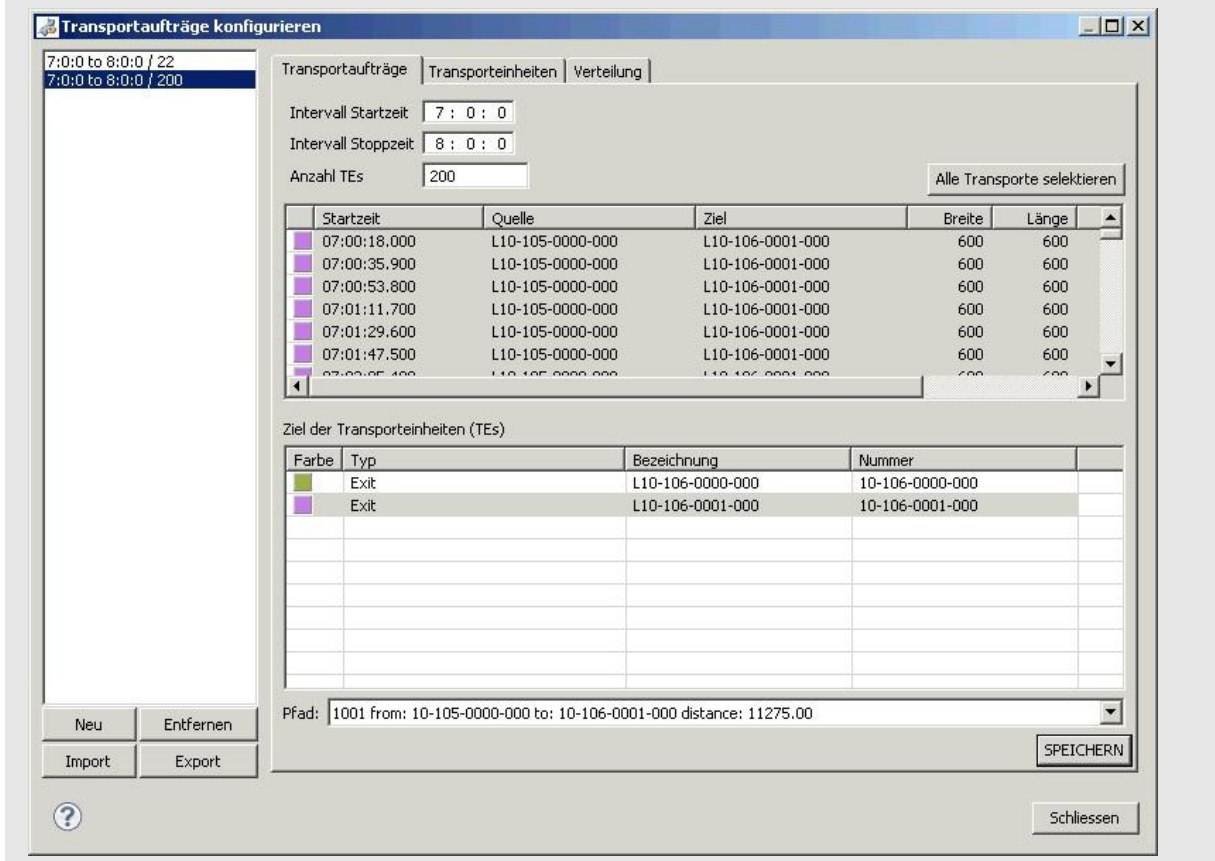
The dialog for defining transport orders opens for racks when you click the “Edit...” button on the left in the element details and then click the “Rules” field.

☐ Elementspezifisch

Anzahl X-Fächer	4
Anzahl Z-Fächer	2
Anzahl Y-Fächer	3
Fachlänge, m	1.00
Fachbreite, m	1.00
Fachhöhe, m	1.00
Zwischenraum (Y), m	0.05
Gestell anpassen	edit...
Lagerbereich	KEINER
3D Sicht	KOMPLEX



Defining Times and Destinations



Top / Left: The definition of 200 shipments between 7:00 a.m. and 8:00 a.m. Additionally, 22 shipments were defined for the same time period. Multiple different order matrices (number of shipments, time windows, and destinations) can be created.

Bottom / Left:

New: Create an additional order matrix

Remove: Delete an order matrix

Import: Importing MS Excel order matrices

Export: Export the selected order matrices to MS Excel

Top / Center:
Start time of the first order (7:00 a.m.), end time (8:00 a.m.), and number of transport orders.

Center:

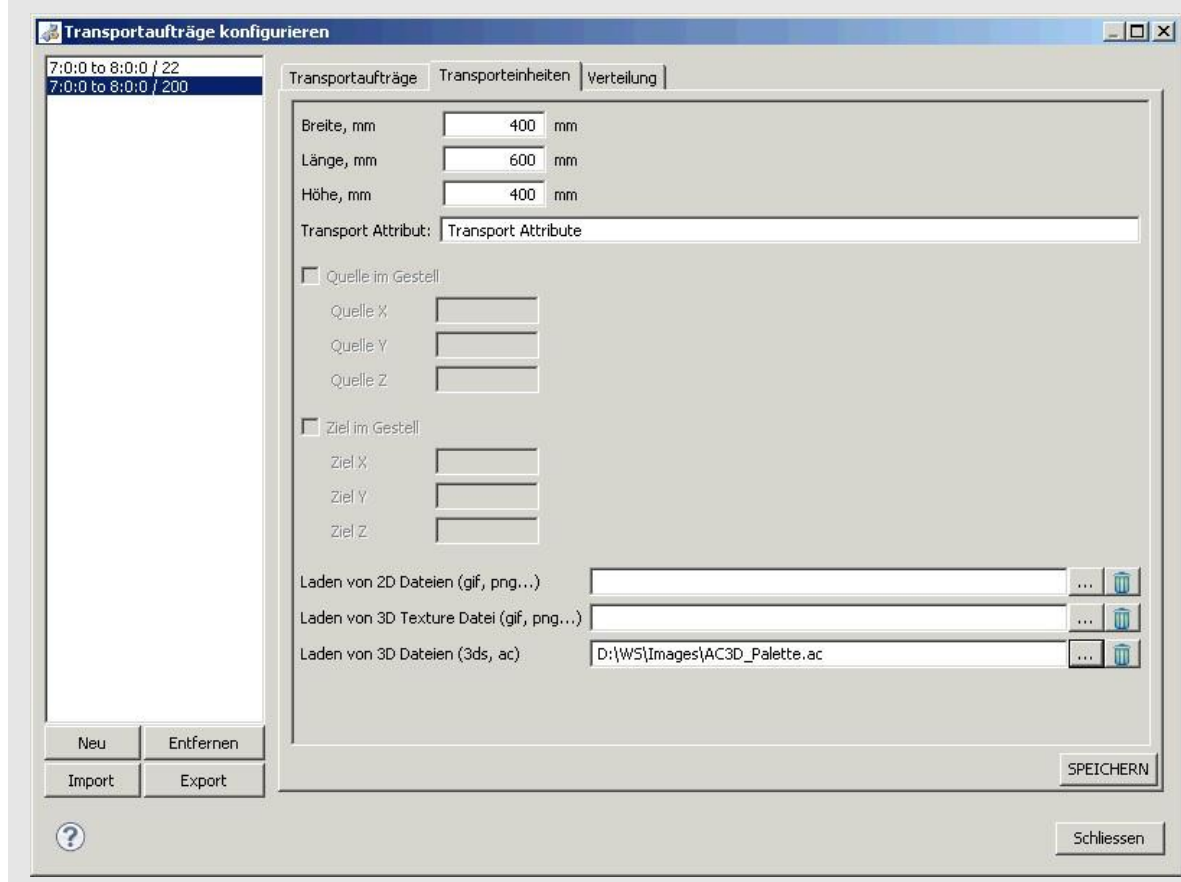
Assigning orders to a destination. Destinations can be assigned to individual orders or to all orders using the "Select All"

"Select All Transports" button.

Center / Bottom:

Selecting a path to the destination. If there are multiple connections to a destination, the path can be selected here. Note: If the layout is adjusted after the path has been selected, the selected path may be lost, or the shortest connection to the destination may become active again (default setting).

Definition of Dimensions and Display of Transport Units



Width, length, and height of the transport unit.

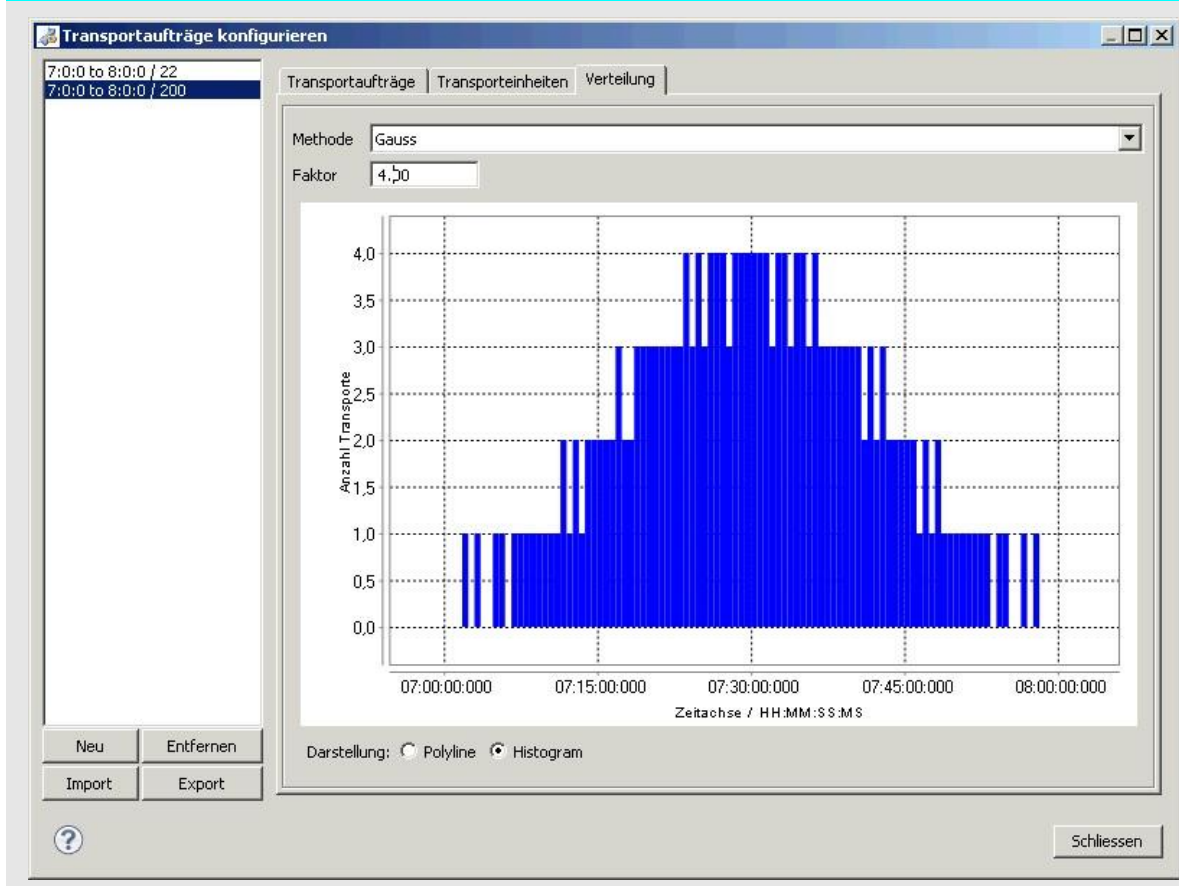
Definition of an attribute.

If the source or destination is a rack, the storage bin can be assigned.

Uploading graphics in the 2D and 3D environments.

Changes must be finalized by clicking the "Save" button.

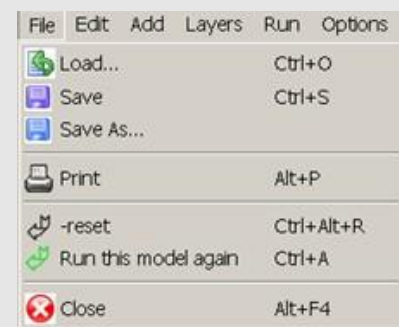
Definition of distributions (exponential, negative exponential, and Gaussian)



Selecting distributions based on the chosen order scenario. The exponential, negative exponential, Gaussian, and linear distributions are supported. Specify the factor and select the display option in the lower section.

VII. Save layout and start

Save and start

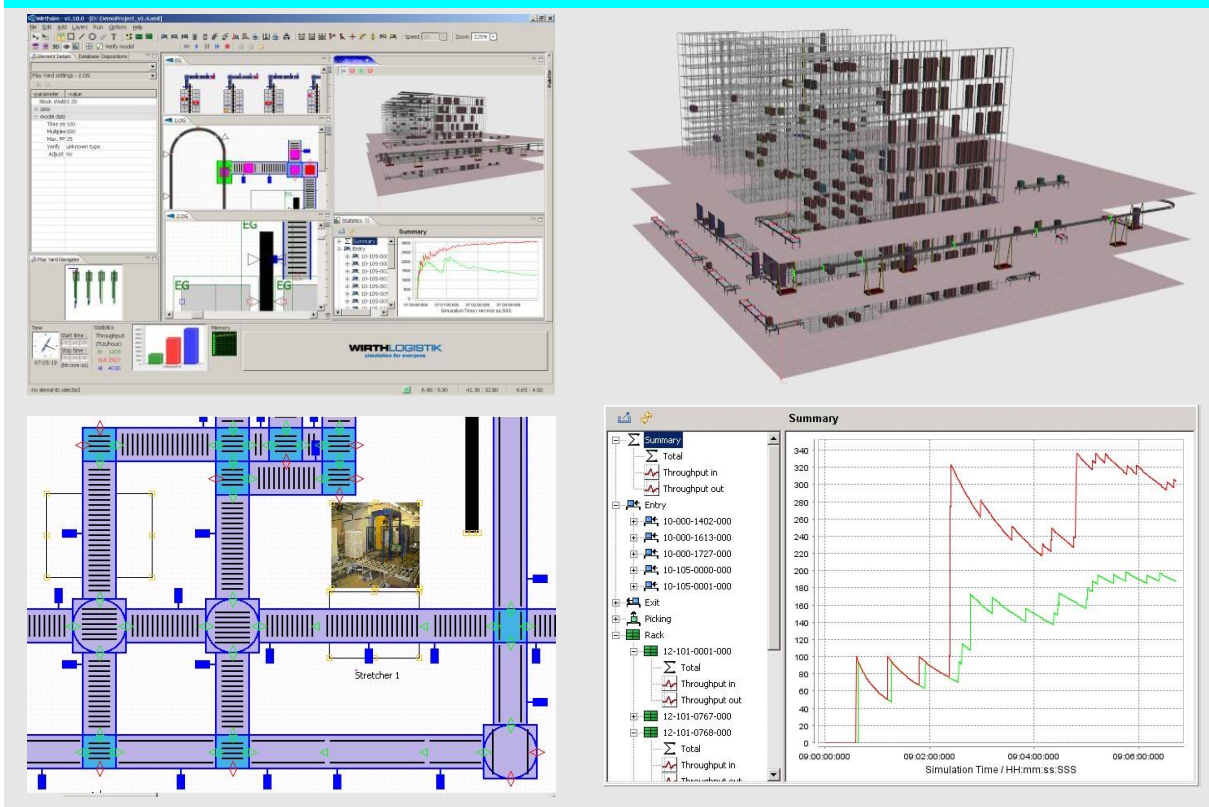


Model Handling



VIII. Monitoring the model in 2D, 3D, and the Statistics dialog

Monitoring of variants / results



9 Basic Settings

The basic settings for each level (floor) are explained below. The settings under “model data” apply across all levels.

Settings in the layers. These are displayed as soon as you click with the mouse on an empty

field. The settings under “Model Data” do not refer explicitly to a specific layer (General Settings).

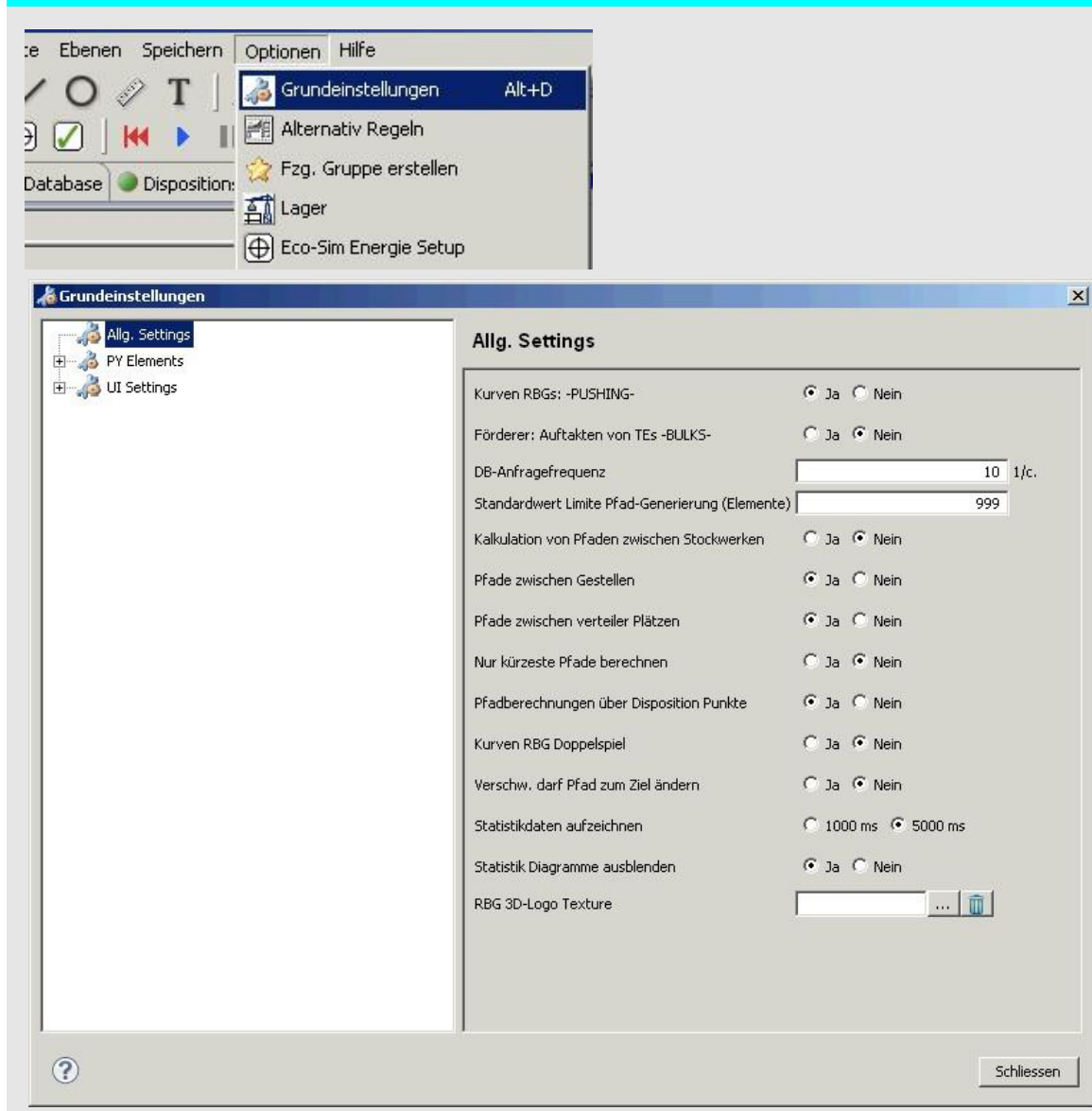
Layer 1	
+ -	
Parameter	Werte
Breite, m	0.20
- sizes	
X-Offset	0.00
Y-Offset	0.00
X-Länge, m	50.00
Y-Länge, m	50.00
Höhe (Ebene), m	10.00
- model data	
Zeitschritte, s (100 = 10x/Sek. / 50 = 20x/Sek.)	100
Multiplikator (1000 = 1x Geschw.)	1000
Max. FPS (Frames pro Sekunde)	25
Verifizieren	unknown type
Max. CPU-Belastung	Nein
Limite Pfad-Generierung (Anz.Elemente)	999

Description			
Width, m. Defines the width of overhead conveyor blocks. No additional functionality.			
X-Offset. Creates an offset from the base coordinate (x0). This applies to the current plane.			
Y-Offset. Creates an offset from the base coordinate (y0). This is relative to the current plane.			
X-Length, m. The extent of the plane relative to the X-coordinate.			
Y-length, m. Extension of the plane along the Y-coordinate.			
Height (plane), m. Height of the corresponding plane.			

Time steps, s. Defines how often the layout is calculated per second. 100 (default setting) => the layout is calculated ten times per second. For example, with a value of 50, the layout is calculated twenty times per second. This ensures that the movement of transport units, RBGs, etc., is displayed smoothly (1x speed). As a result, the computational load doubles, and the duration of a simulation run—for example, one lasting 5 hours—is extended. For example, with a value of 200, the quality of the display of movements for transport units, RBGs, etc., deteriorates. This is because the layout is calculated only five times per second. The duration of a simulation run is significantly shortened. If this value is increased further, the functionality of the created layout can no longer be guaranteed in all cases!			
Multiplier. This value corresponds to the “Speed” setting. Geschwindigkeit 1x 			
Max. FPS (frames per second). Interval for rendering in the 3D environment.			
Max. CPU load. Ensures that a simulation run completes as quickly as possible. All available resources on a computer are allocated to WirthSim.			
Path Generation Limit. Defines the maximum number of elements from a source to a destination that are taken into account. If this value is set to three, for example, a maximum of three elements—such as conveyors, RBGs, etc.—are considered when calculating a path. This value is limited accordingly for more complex layouts and, in particular, for EHB systems. This value is saved along with the layout and taken into account when loading (calculating the paths).			

Default settings under “Options/Default Settings.”

Basic Settings

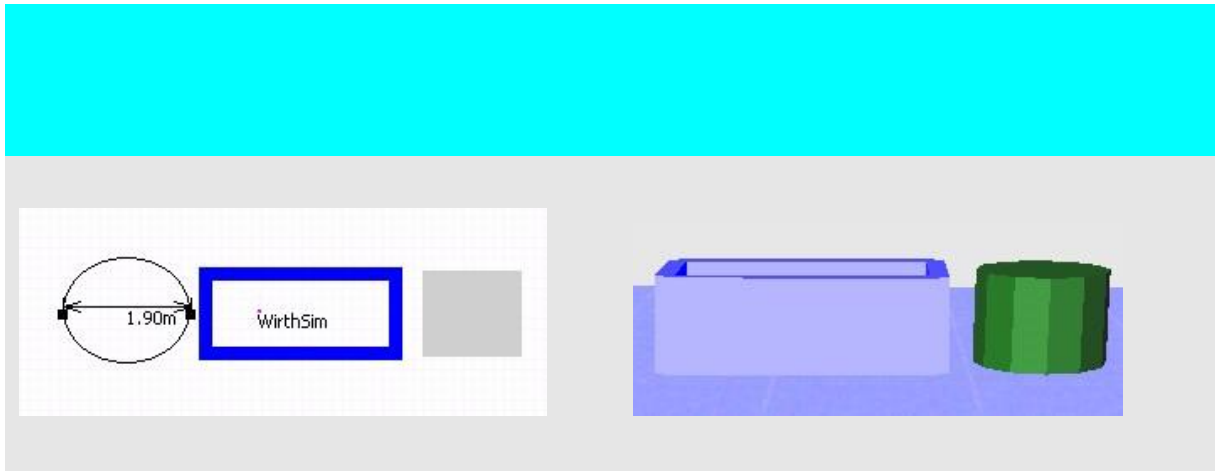


Description			
Curves RBGs –PUSHING–. Defines whether RBGs or shuttles should avoid each other when RBGs or vehicles are blocking one another in the vicinity of curves.			

TE Start-up –BULKS–. NO: On a conveyor element with multiple stop positions, synchronizing occurs only if the downstream conveyor is empty. If “YES,” synchronizing always occurs.			
DB Query Frequency. Defines a ratio between the simulation processing cycles and the database interface. If the value is set to 1 1/c, a the database is accessed during the processing of every simulation cycle. At 10 1/c, this occurs only every tenth cycle.			
Path Generation Limit (No. of Elements). Limits the number of elements to be considered when generating a path (route => from a source to a destination). This setting is used in the context of overhead monorail systems and large logistics systems to avoid calculating unnecessary paths.			
Generate paths through RBGs (floors). Enables or disables the calculation of paths between floors. No paths (routes) from one floor to another via rack-servicing machines should be calculated.			
Generate paths between racks. Defines whether paths between racks are calculated.			
Generate paths between distributors. Conveyor elements can be defined as part of a distributor under the “Element-specific” setting. This setting enables or disables the calculation of paths between distributors.			
Calculate only the shortest paths. If this setting is enabled, only the shortest connection from a source to a destination is calculated. Enabling this setting significantly reduces the loading time for larger layouts.			

Allow paths via routing points. Conveyor elements include a setting under “Routing” (Routing Point) for defining an intermediate destination. Paths are calculated up to these destinations and simultaneously across them. If this setting is set to “No,” no paths will be calculated “OVER” this point.			
Allow double-cycle RBG curves. Forces the next transport unit (RBG) to be dispatched from the same rack after a transport unit has been dispatched. This breaks the order FIFO sequence.			
The transfer cart may change its path to the destination. If the			
drop-off position of a transfer car is occupied and another path to the transport unit's destination exists, this setting allows the route (path) of a transport unit on the transfer car to be adjusted.			
Record statistical data. Interval for recording statistical data.			
Hide statistics charts. Processing the data, including timestamps, for the charts within the statistics dialog consumes a significant amount of application resources. This is undesirable for more complex layouts.			
RBG 3D Logo Texture. A customer logo, etc., can be loaded and is displayed on the mast of the RBG.			

10 General Elements

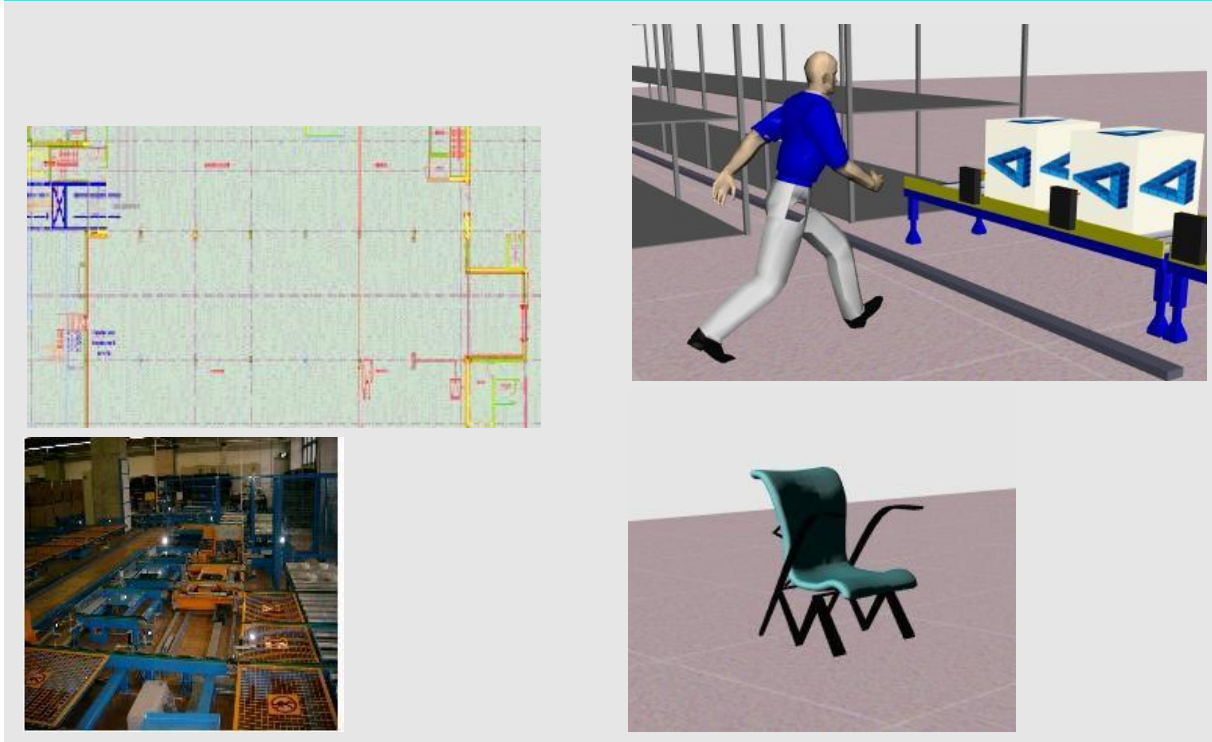


These buttons and functions are used to control the application and configure environment variables.

10.1 "Image" Element

2D Development View

3D View



The Image element supports the display of both 2D and 3D representations within the simulation. 2D images can be loaded directly from the hard drive into the WirthSim application or placed in the layout. 3D elements can be selected from the internal WirthSim library or imported in *.3ds or *.ac format.

Examples of use include:

- ▶ As a 2D background image. For example, layout plans of buildings and installations.
- ▶ 2D images of real-world systems, such as company logos.
- ▶ 3D images for representing walls, tables, people, etc. ▶ For representing machines or machining centers

Images are important for establishing a connection between the virtual (simulation) and the real environment.

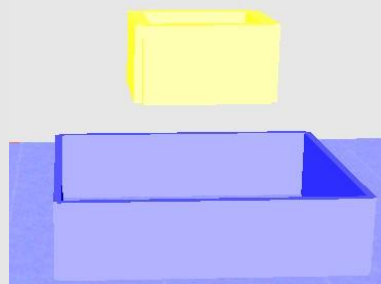
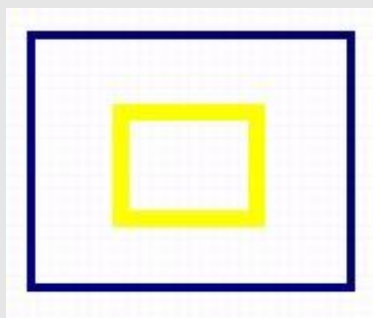
The following 2D image formats are supported: *.jpg, *.gif, *.png, *.bmp, and *.ico. 3D image formats are: *.3ds and *.ac (AC3D => <http://www.inivis.com/>)

Description			
Element supported in:			
Variable length, width, and angle			
Variable height (used only for 3D rendering)			
Offset from the ground (used only for 3D display)			
Loading from the hard drive and saving to the project XML file.			
Image is in the background: yes/no			
Image is locked (cannot be moved while layout)			
Selecting 3D images from the library			

10.2 Quadrilateral Element

2D Development View

3D view



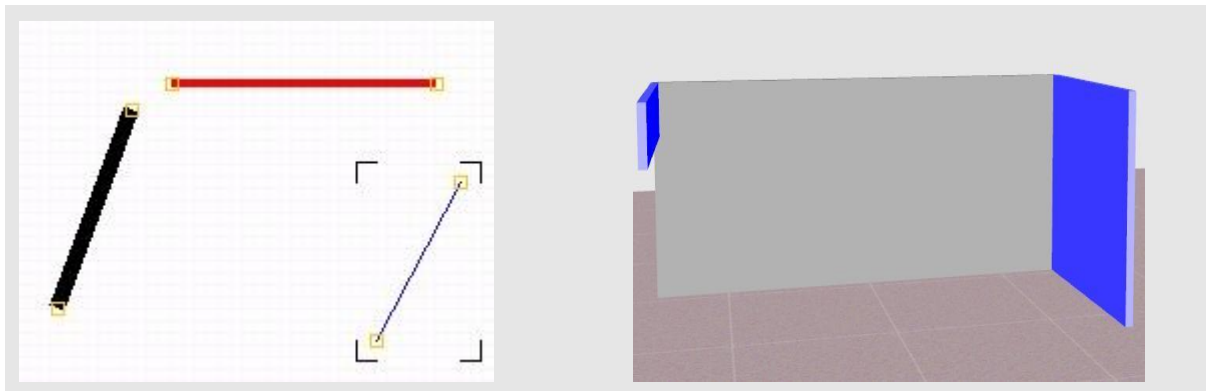
Used to display rectangles. 3D rendering is also supported.

Description			
Element supported in:			
Variable length, width, and angle			
Variable height (used only for 3D rendering)			
Offset from the ground (used only for 3D display)			
Variable line width			
Color Selection			
Image in the background: yes/no			

10.3 Line Element

2D Development View

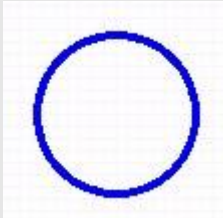
3D view



Used to display lines. 3D display is also supported.

Description			
Element is supported in:			
Variable length, width, and angle			
Variable height (used only for 3D rendering)			
Offset from the ground (used only for 3D display)			
Variable line width			
Color Selection			
Image in the background: yes/no			

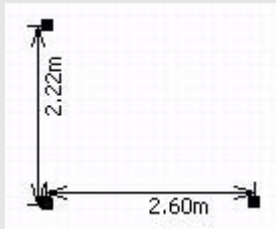
10.4 Circle Element

2D Development View	3D view
	Not supported

Used to display circles. Display is not supported in 3D.

Description			
Element is supported in:			
Variable length and width			
Variable line width			
Color Selection			
Image in the background: yes/no			

10.5 Reference Element

2D development view	3D view
	Not supported

Used for measuring distances. It establishes a reference between distances within the virtual system (simulation) and the real environment. 3D display is not supported.

Description			
Element supported in:			
Variable Length			
Angle			
Distance displayed above or below the line.			
Color selection			

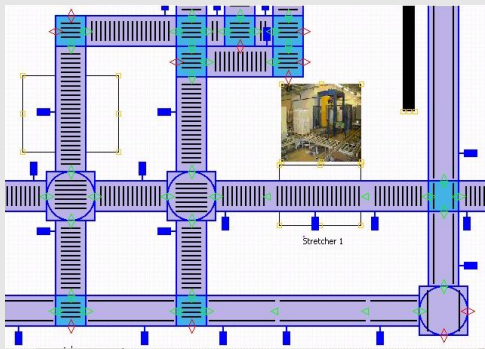
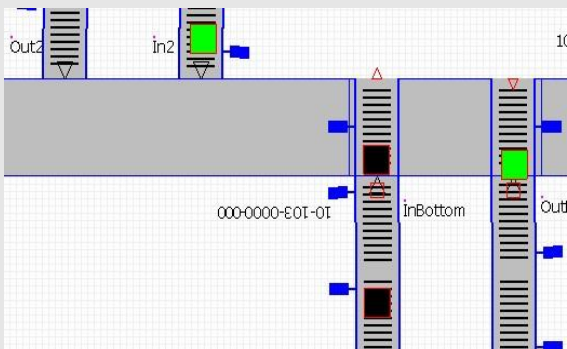
10.6 Text Element **T**

2D Development View	3D view
	Not supported

Used for creating and displaying text. Display is not supported in 3D.

Description			
Element is supported in:			
Variable angle			
Text field			
Color Selection			

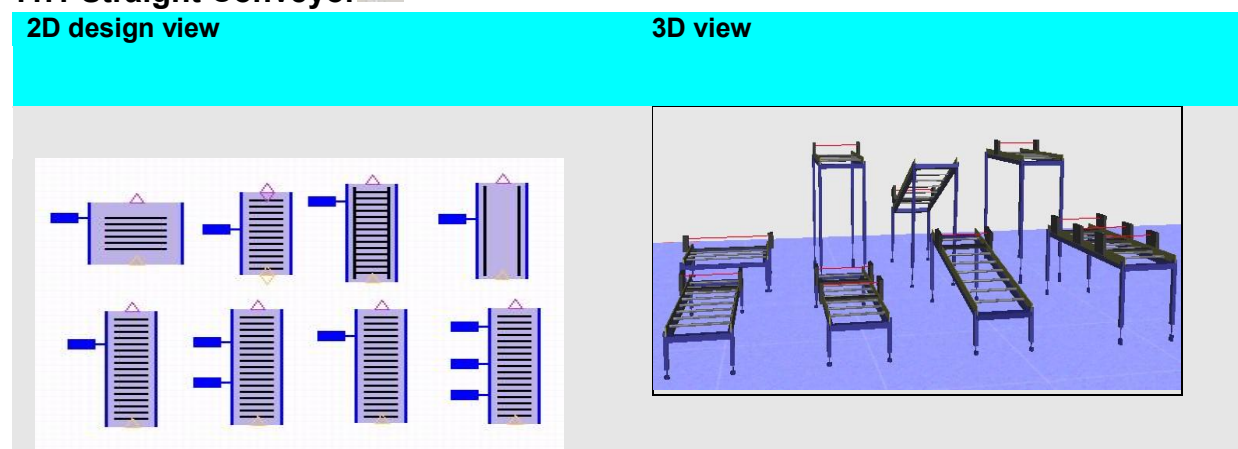
11 Conveyor Elements

2D Development View	3D view
	

Conveyor elements are used for the automatic transport of goods. Such goods are transported on or in transport units (TUs) such as boxes, pallets, and containers

conveyors. These can be light (e.g., a carton) or heavy (e.g., a pallet). To create a simulation of a material flow based on conveyors, different element types are used. Conveyor systems for light transport units, such as cartons or containers, feature a higher transport speed and high throughput. In this environment in particular, it is important to take acceleration and deceleration ramps into account. WirthSim supports individual acceleration and deceleration ramps for each conveyor element.

11.1 Straight Conveyor



This conveyor element can be configured with 1 to n stop positions. Each stop position corresponds to a transport unit on this element. It supports accumulation, de-accumulation, and group transport. The number of stop positions determines whether the conveyor feeds in, feeds out, or transports a group of transport units.

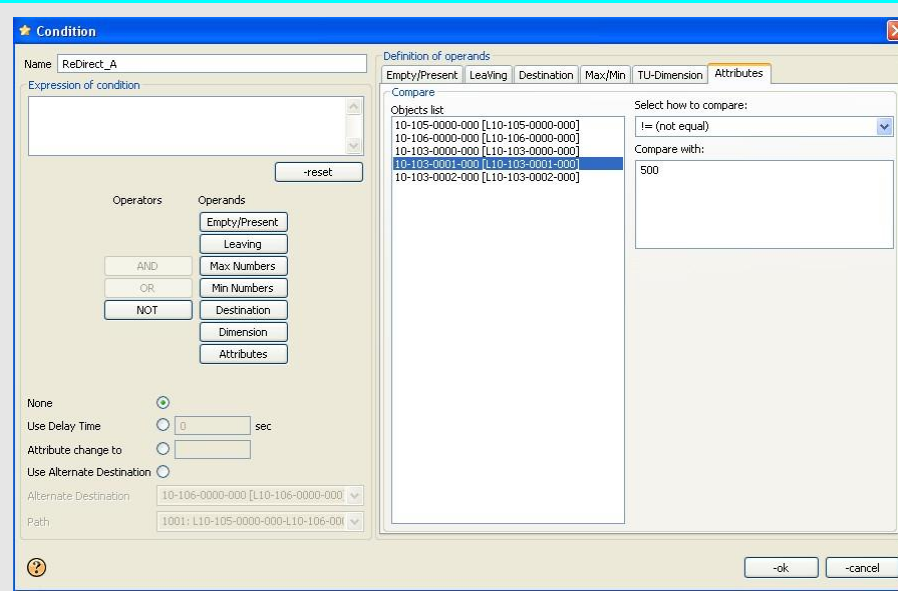
This element is always used when the conveyed goods must not come into contact with one another. Its main applications include: pallet conveyor systems, mechanical engineering, and container conveyor systems.

Description			
Element is supported in:			
Variable length, width, and angle			
Element configuration as a roller, chain, or belt conveyor			
Display of element supports: YES/NO			
Variable height (north/south connection point)			

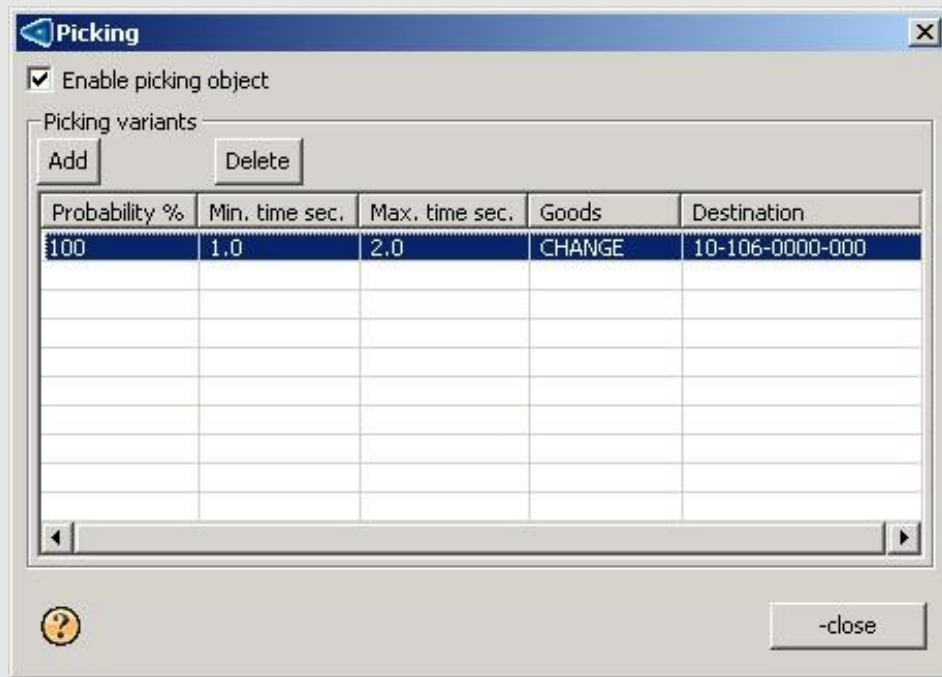
Variable acceleration (m/s^2), braking ramp (m/s^2), and speed (m/s) per conveyor element			
Bidirectional motion			
Variable Delay Time			
Option to define this element as a scheduling target			
MRP: PRA Setting (Algorithm for putting away and removing items from racks)			
Scheduling: Update / Request Settings interaction / external scheduling			
Scheduling: Definition of Dependencies			
MRP: Define element as a scheduling point			
Max. number of TEs en route to this scheduling point or Comm. element			
Eco-Sim Energy consumption (kW)			

11.1.2 Dialog Boxes

Define Dependencies (Conditions)



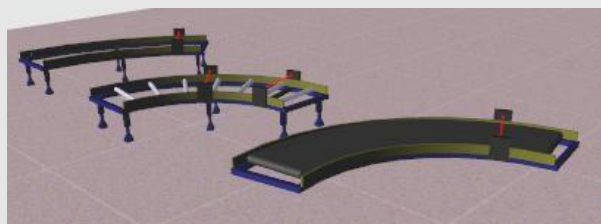
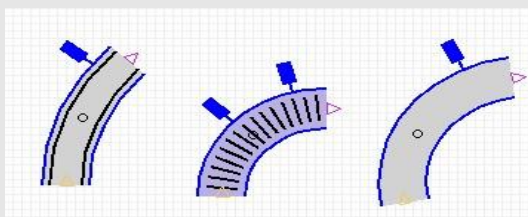
Definition as a Distributor



11.2 Curved Conveyor

2D Development View

3D view



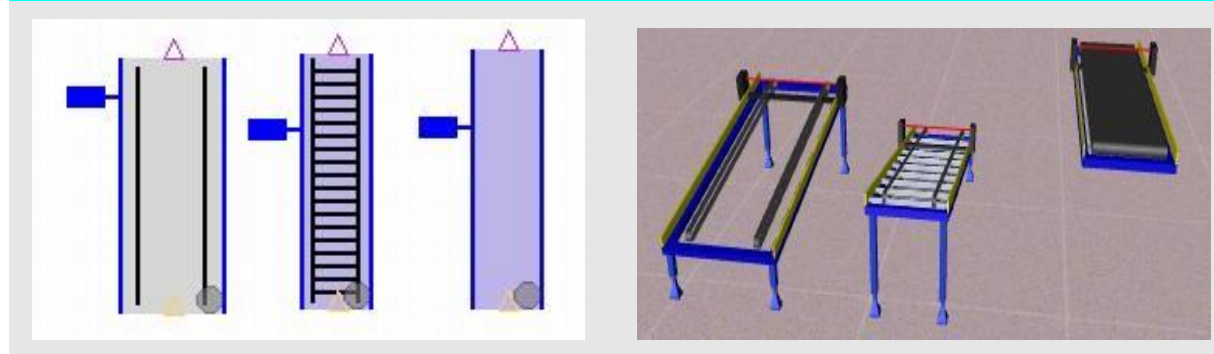
This conveyor element can be configured with 1 to n stop positions. Each stop position corresponds to a transport unit on the conveyor. It supports accumulation, de-accumulation, and group transport. The number of stop positions used determines whether the conveyor operates in a synchronized, de-synchronized, or group transport mode.

Description			
Element is supported in:			
Variable length, width, and angle			
Element configuration as a roller, chain, or belt conveyor			
Display of element supports YES/NO			
Definition of the curve radius / angle			
Variable height (no slope possible)			
Variable acceleration (m/s^2), deceleration ramp (m/s^2), and speed (m/s) per conveyor element			
Bidirectional motion			
Variable Delay Time			
Option to define this element as a scheduling target			
MRP: PRA Setting (Inbound and Outbound Algorithm for crane positions)			
Scheduling: Update / Request Settings Interaction / External Scheduling			
Scheduling: Definition of Dependencies			
Scheduling: Element defined as a distributor.			
Max. number of TEs en route to this scheduling point or communication element			
Eco-Sim Energy consumption (kW)			

11.3 Straight Accumulation Conveyor

2D development view

3D view



This conveyor element supports a maximum of one stop position and simulates a mechanical stop that holds back or releases the transport units. The backpressure can be defined using a parameter (maximum number of transport units in this element). It supports accumulation and de-accumulation (no group transports).

This element is often used in cardboard box and container conveyor systems.

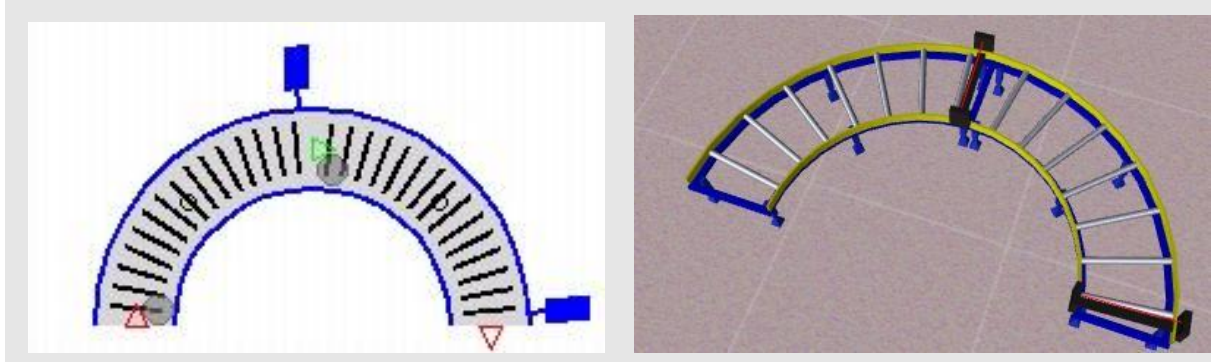
Description			
Element is supported in:			
Variable length, width, and angle			
Variable height (north/south connection point)			
Element configuration as a roller, chain, or belt conveyor			
Display of element supports: YES/NO			
Variable acceleration (m/s^2), deceleration ramp (m/s^2), and speed (m/s) per conveyor element			
Maximum number of transport units on this element.			

Variable delay time			
Configuration: PRA Setting (Algorithm for placing and removing crane positions)			
Scheduling: Update / Request Settings interaction / external scheduling			
Scheduling: Definition of Dependencies			
Eco-Sim Energy consumption (kW)			
Eco-Sim Coast-down time (sec.)			

11.4 Curved Accumulation Conveyor

2D Development View

3D view



This conveyor element supports a maximum of one stop position and simulates a mechanical stop that holds back or releases the transport units. The back pressure can be

defined using a parameter (maximum number of transport units in this element). It supports accumulation and de-accumulation (no group transports).

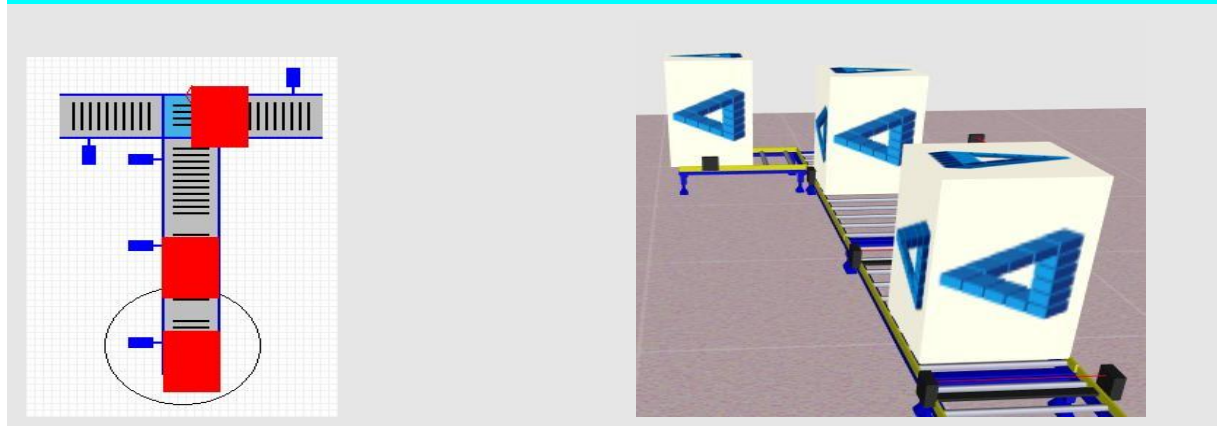
This element is often used in cardboard box and container conveyor systems.

Description			
Element is supported in:			
Variable length, width, and angle			
Definition of the curve radius / angle			
Variable height (no slope possible)			
Element configuration as a roller, chain, or belt conveyor			
Display of element supports: YES/NO			
Variable acceleration (m/s^2), deceleration ramp (m/s^2), and speed (m/s) per conveyor element			
Maximum number of transport units on this element.			
Variable delay time			
Option to define this element as a scheduling target			
Scheduling: PRA Setting (Algorithm for putting away and retrieving items at the crane location)			
Scheduling: Update / Request Settings interaction / external scheduling			
Scheduling: Definition of Dependencies			
Eco-Sim Energy consumption (kW)			
Eco-Sim Coast-down time (sec.)			

11.5 Smuggler

2D development view

3D view



The Feeder element is used as the starting point for transports that are to be fed into the system (model). This conveyor element can be configured with a maximum of one stop position. It does not support accumulation, de-accumulation, or group transports.

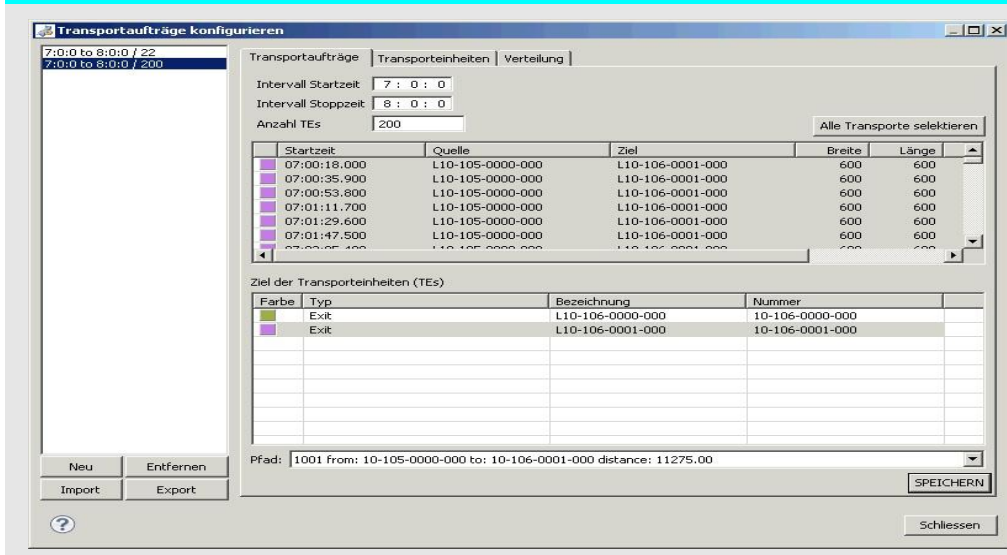
This element is used for all types of conveyed material.

Description			
Element is supported in:			
Variable length, width, and angle			
Variable height (north/south connection point)			
Element configuration as a roller, chain, or belt conveyor			
Display of element supports YES/NO			

Variable acceleration (m/s^2), deceleration ramp (m/s^2), and speed (m/s) per conveyor element			
Handling of transport orders / Generation			
Variable Delay Time			
Scheduling: Definition of Dependencies			
Eco-Sim Energy consumption (kW)			

11.5.2 " " dialog boxes

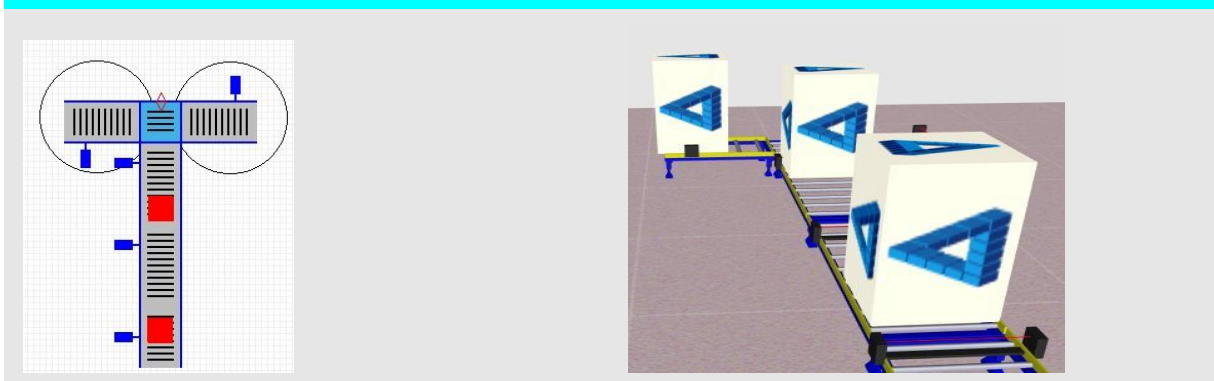
Definition of Transport Orders



11.6 Ejector

2D Development View

3D View



The output conveyor element is used as the destination for transports that are to be discharged from the system (model). This conveyor element can be configured with a

maximum of stop positions. It does not support accumulation, de-accumulation, or group transports.

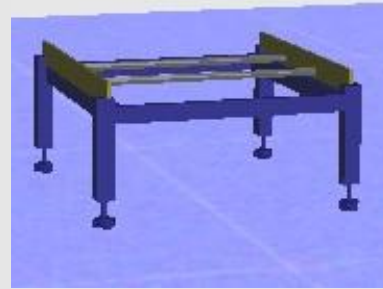
This element is used for all types of conveyed goods.

Description			
The component supports:			
Variable length, width, and angle			
Variable height (north/south connection point)			
Element configuration as a roller, chain, or belt conveyor			
Display of element supports: YES/NO			
Variable acceleration (m/s^2), deceleration ramp (m/s^2), and speed (m/s) per conveyor element			
Variable deceleration time until the conveyor is empty			
Configuration: Element defined as a distributor.			
Maximum number of TEs en route to this dispatcher			
Eco-Sim Energy consumption (kW)			

11.7 Stroke Converter - Pusher

2D design view

3D view



The transfer unit, or pusher, stops transport units at the center of the element. Up to four elements can be connected to it. It supports a starting position at the top or bottom, as well as a time for lifting or lowering.

This element is used at corners and intersections within pallet systems. It can also be used as a pusher within light-goods transport systems. If used as a pusher, the push speed, push acceleration, and push braking ramp can be defined for this application. Please note that these definitions must account for the pusher's return movement to its home position.

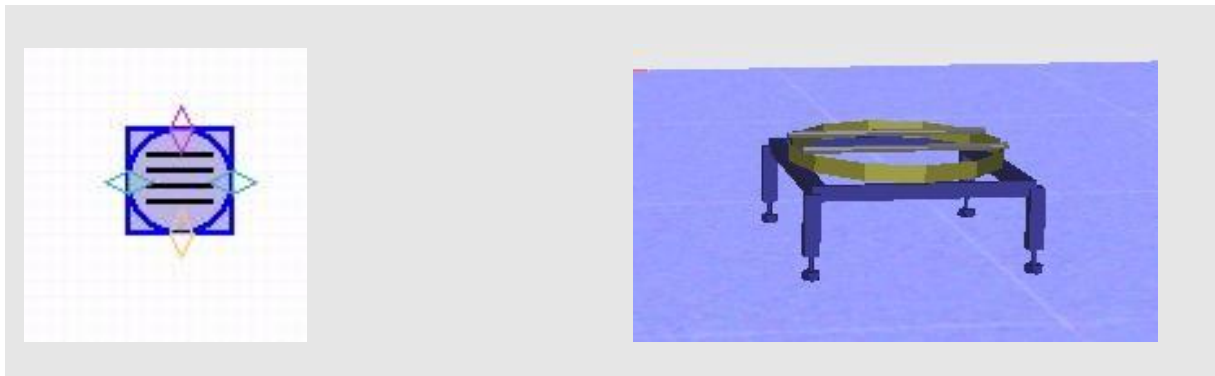
Description			
Element is supported in:			
Variable width and angle			
Variable Element Height			
Element configuration as a roller, chain, or belt conveyor			
Display of element supports: YES/NO			

Acceleration (m/s^2), deceleration ramp (m/s^2), and speed (m/s) per conveyor element			
Definition of the starting position (top/bottom)			
Increase/decrease time			
Delay time			
Option to define this item as a scheduling target			
MRP: PRA Setting (Algorithm for putting away and removing items from racks)			
Scheduling: Update / Request Settings Interaction / External Scheduling			
Scheduling: Definition of Dependencies			
Scheduling: Element defined as a distributor.			
Max. number of TEs en route to this dispatch point or communication element			
Eco-Sim Energy consumption (kW)			

11.8 Rotary Table

2D development view

3D view



The turntable stops transport units at the center of the element. Up to four elements can be connected to it. It supports a starting position (default position) as well as a rotation time.

This element is used in corners and at intersections, primarily in pallet systems.

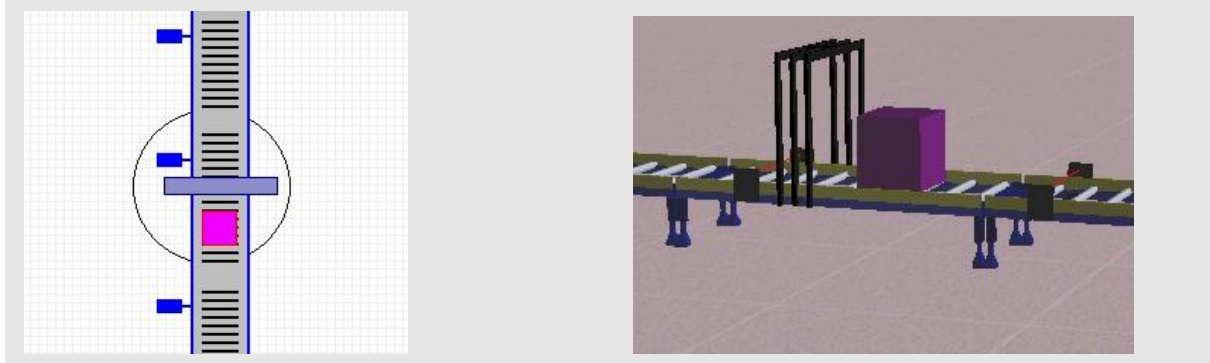
Description			
Element is supported in:			
Variable width and angle			
Variable Element Height			
Element configuration as a roller, chain, or belt conveyor			
Display of element supports			
Acceleration (m/s^2), deceleration ramp (m/s^2), and speed (m/s) per conveyor element			
Definition of the starting position (alignment)			
Direction of rotation (clockwise or counterclockwise—90 or 270 degrees)			
Time required for the rotation (90 degrees)			
Delay time			

Option to define this item as a scheduling target			
MRP: PRA Setting (Inbound and Outbound Algorithm for crane positions)			
Scheduling: Update / Request Settings interaction / external scheduling			
Scheduling: Definition of Dependencies			
Scheduling: Element defined as a distributor.			
Max. number of TEs en route to this availability point or communication element			
Eco-Sim Energy consumption (kW)			

11.9 Profile Check

2D development view

3D view



This conveyor element is used for quality control of the conveyed goods. This includes length, height, width, etc. If the conveyed goods—such as cardboard boxes, containers, or pallets—do not

requirements, it is subsequently diverted out of the system for inspection. It does not support accumulation, de-accumulation, or group transport.

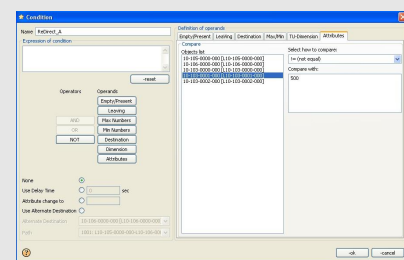
Almost every automated transport system has an input inspection. This is to ensure high system availability at all times.

Description			
Element is supported in:			
Variable length, width, and angle			
Variable height (north/south connection point)			
Element configuration as a roller, chain, or belt conveyor			
Display of element supports: YES/NO			
Variable acceleration (m/s^2), deceleration ramp (m/s^2), and speed (m/s) per conveyor element			
Variable deceleration time			
Option to define this element as a scheduling target			
MRP: PRA Setting (Inbound and Outbound Algorithm for crane positions)			
Scheduling: Update / Request Settings interaction / external scheduling			
Scheduling: Definition of Dependencies			
Scheduling: The number of dispatches can be defined.			

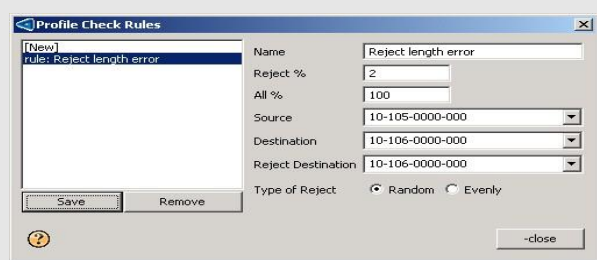
Eco-Sim			
Energy consumption (kW)			

11.9.2 " " dialog boxes

Defining Dependencies



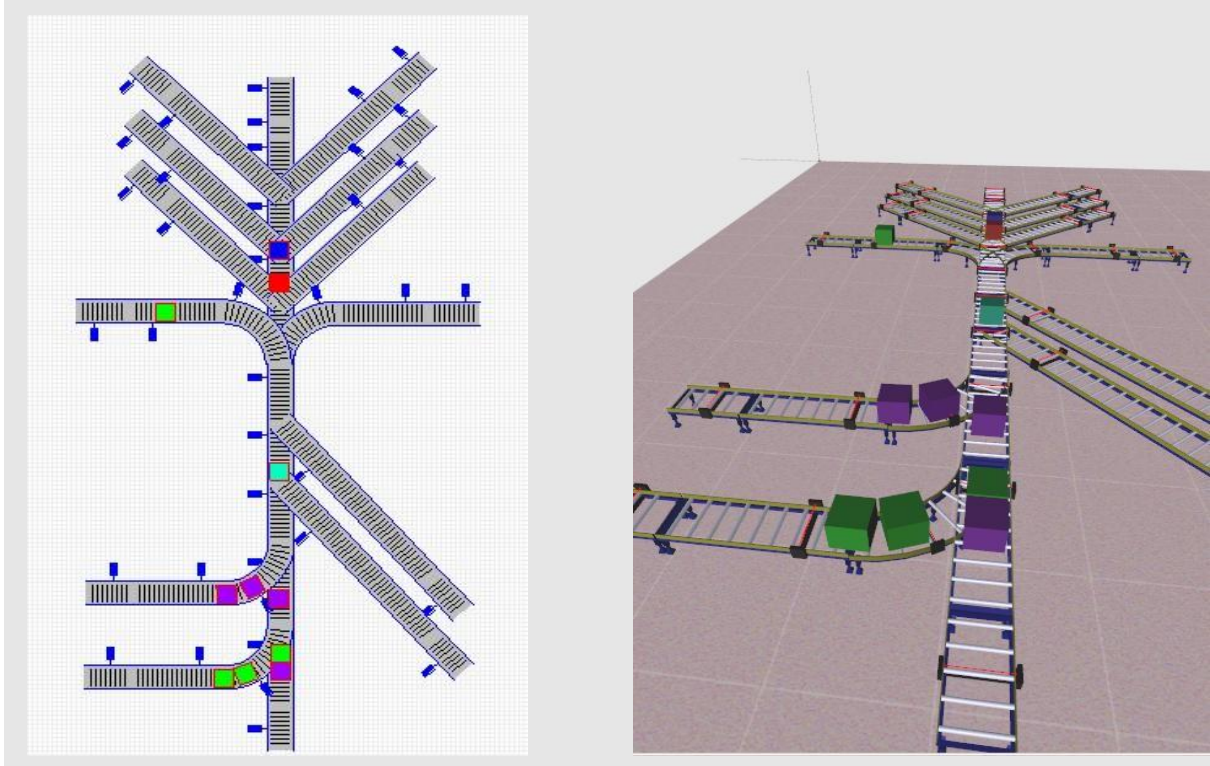
Release Rules



11.10 45-Degree Infeed and Outfeed

2D Development View

3D view



The 45-degree infeed and outfeed elements can be created using straight lines as well as curved elements. The infeed and outfeed behavior can be controlled using the standard rules (Alternating / Conditions). Connecting conveyor elements to these types is only possible in the Professional version.

Description

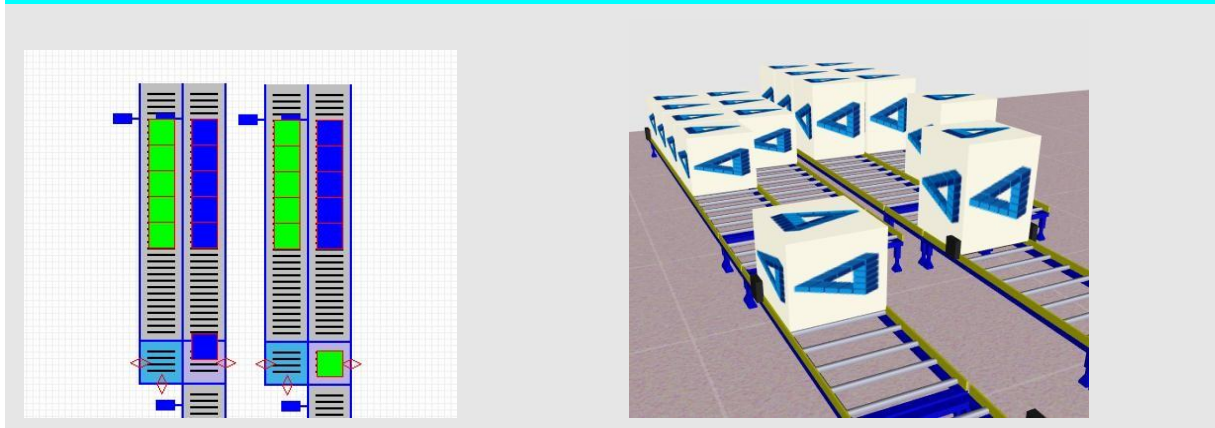
Element supported in:

These elements are created from the funding types described above.

11.11 Truck Dock

Development view 2D

3D view



The truck dock is used as a destination for transports that are to be discharged from the system. This conveyor element can be configured with a maximum of one stop position.

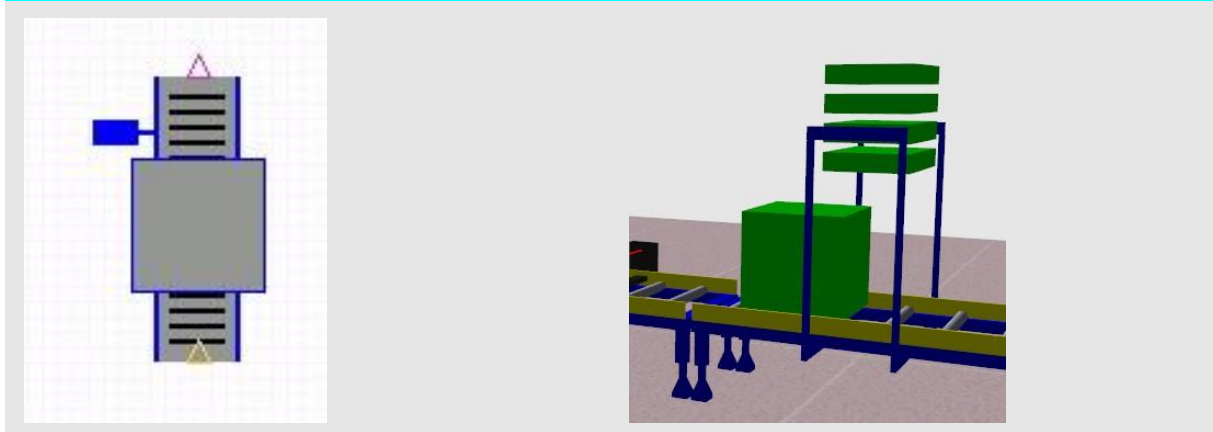
This element is used for the automatic loading of trucks.

Description			
Component supported in:			
Variable length, width, and angle			
Variable height (north/south connection point)			
Element configuration as a roller, chain, or belt conveyor			
Display of element supports: YES/NO			
Variable acceleration (m/s^2), deceleration ramp (m/s^2), and speed (m/s) per conveyor element			
Variable deceleration time until the conveyor is empty			
Maximum number of transport units			
Max. number of transport units en route to this scheduling point or communication element			
Eco-Sim			
Energy consumption (kW)			

11.12 (Un)stacker

2D development view

3D view



The (De)Stacker element enables the stacking and destacking of TEs. Based on the destination, you can specify whether TEs simply pass through the (De)Stacker element without being stacked or whether they are stacked. A dialog box allows you to customize the element's behavior.

Furthermore, it is possible to define a "1-to-n" or "n-to-1" relationship. This allows TEs to be "split" or merged into a single TE.

This element is typically used for stacking pallets, in pallet dispensers, and within container conveyor systems.

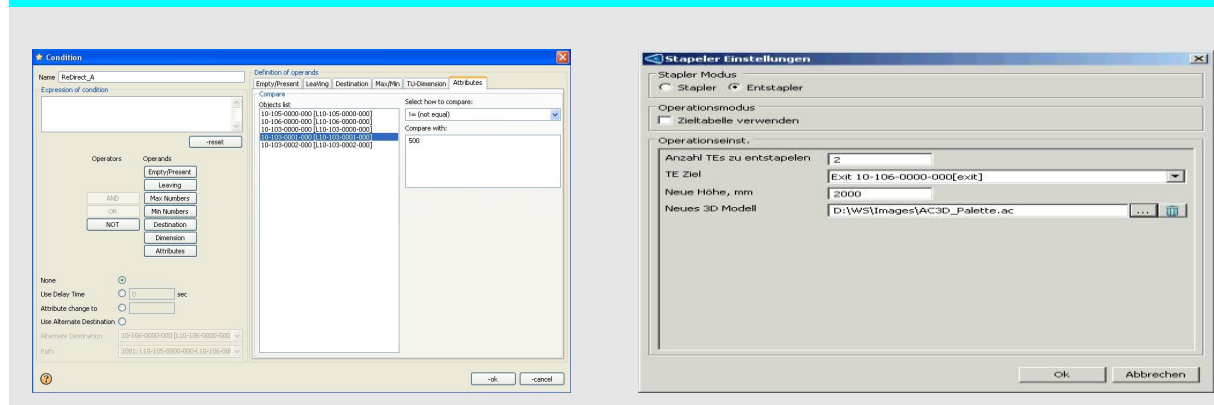
Description			
Element is supported in:			
Variable length, width, and angle			
Variable height (north/south connection point)			
Variable acceleration (m/s^2), deceleration ramp (m/s^2), and speed (m/s) per conveyor element			
Variable deceleration time			
Option to define this element as a scheduling target			
Specifying the height of the TE (when it leaves the forklift)			

Storing the new 3D graphic (when the TE leaves the forklift)			
Scheduling: Update / Request Settings interaction / external scheduling			
Scheduling: Definition of Dependencies			
Scheduling: Specify whether this element is used as a stacker or a destacker.			
Eco-Sim Energy consumption (kW)			

11.12.2 Dialogs

Define Dependencies

Purpose of the batch



11.13 Lift

2D Development View

3D view



The elevator element connects different floors or levels within floors. It can accommodate n-TEs and transport each one to the corresponding floors.

Variants:

- ▶ Input only (transport units—TEs—are loaded)
- ▶ Input and Output (Transport Units (TEs) are conveyed into and out of the elevator)
- ▶ Exit only (transport units—TEs—can only leave the lift)
- ▶ Two Inputs (Transport Units (TEs) can enter the elevator from both sides)

Description			
Element is supported in:			
Variable width and angle			
Variable Element Height (Conveyor)			
Variable Height (North/South Connection Point)			
Element configuration as a roller, chain, or belt conveyor			
Acceleration (m/s^2), deceleration ramp (m/s^2), and speed (m/s) of the conveyor element			
Lift cycle times according to defined vertical parameters			
Rest position of the hoist (floor / platform)			
Lifter operating modes: pick up only, pick up and set down, and set down only			
Delay time			
Option to define this item as a scheduling target			
MRP: PRA Setting (Inbound and Outbound Algorithm for crane positions)			
Scheduling: Update / Request Settings interaction / external scheduling			
Scheduling: Definition of Dependencies			
Eco-Sim Energy Consumption of Equipment (kW)			
Braking Energy Consumption (kW)			
Energy consumption at v-max. (kW)			

11.13.2 Dialogs

Define Dependencies

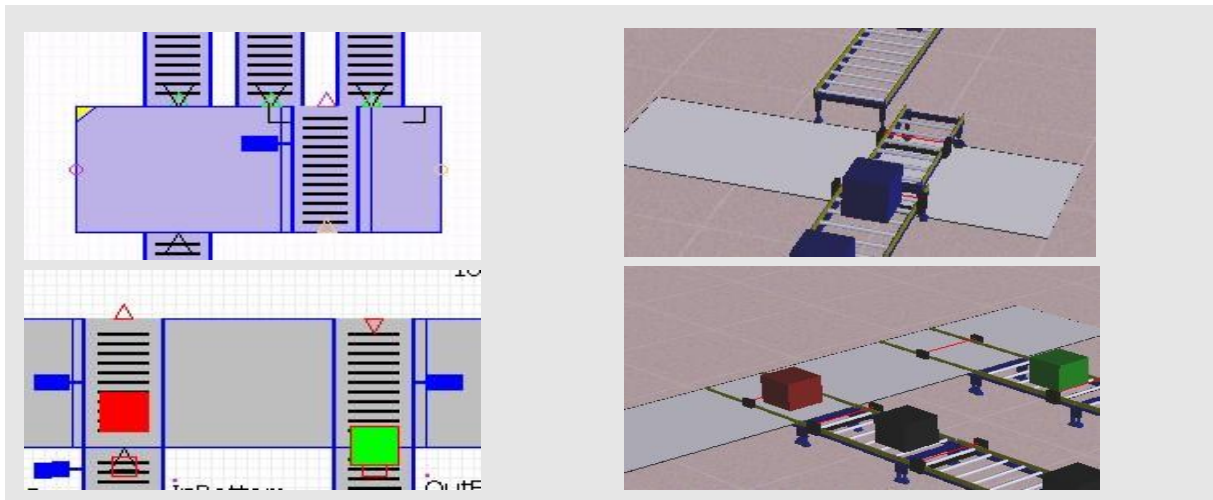
Defining the routes between floors or landings

Size	
Block Length	2.80
Block Width	0.80
Block Angle	90.00
3D Height	1.00
Height	0.50
Speed	
Max. speed	1.00
Acceleration ramp	0.00
Deceleration ramp	0.00
Max_Vertical_Speed	1.00
Vertical_Acceleration	0.00
Vertical_Deceleration	0.00

11.14 Distribution Cart

2D development view

3D view



The distribution cart connects multiple conveyor lines to one another. A start position (default position) can be defined. The conveyor(s) on this element support bidirectional mode. The distribution cart supports the following applications:

- ▶ Conveyors with one stop position (for a pallet, container, etc.)
- ▶ Conveyors with two stop positions (for two pallets, containers, etc.) ▶

Multiple dual-conveyor variants

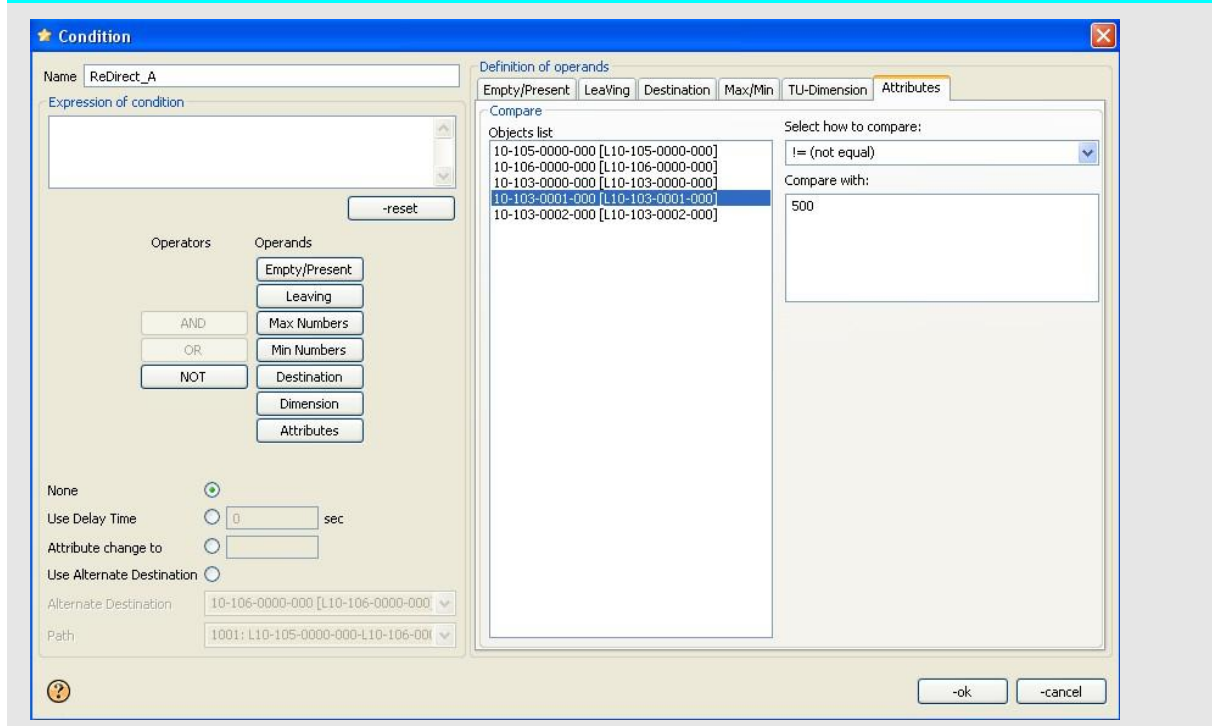
The distribution cart is used for both pallet and container systems. It is generally not used in system sections with high throughput.

Description			
Element supports:			
Variable-width and variable-angle transfer carts			
Variable Conveyor Width			
Variable Element Height (Conveyor)			
Variable Height (North/South Connection Point)			
Element configuration as a roller, chain, or belt conveyor			
Acceleration (m/s^2), deceleration ramp (m/s^2), and speed (m/s) of the distribution cart			
Acceleration (m/s^2), deceleration ramp (m/s^2), and speed (m/s) of the conveyor element			

Definitions when two conveyors are used, such as distance....			
Default position of the distribution cart			
Delay time			
Option to define this item as a scheduling target			
MRP: PRA Setting (Algorithm for putting away and removing items from racks)			
Scheduling: Update / Request Settings Interaction / External Scheduling			
Scheduling: Definition of Dependencies			
Selection of the operation variant (one or two conveyors, etc.)			
Eco-Sim Energy consumption of the system (kW)			
Braking energy consumption (kW)			
Energy consumption at v-max. (kW)			

11.14.2 Dialogs

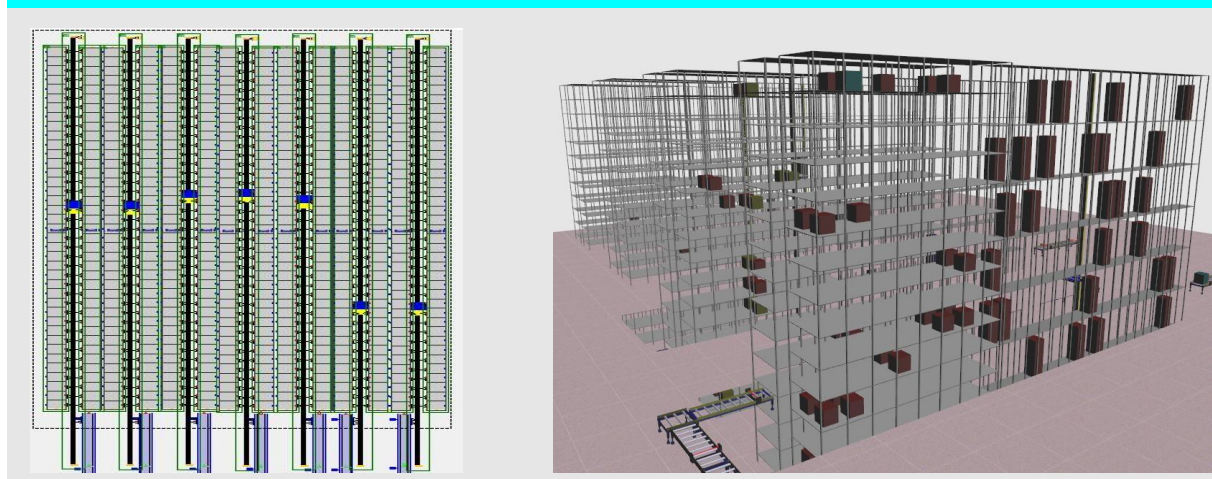
Define Dependencies



12 storage and retrieval machines, shuttles, and racks

2D design view

3D view



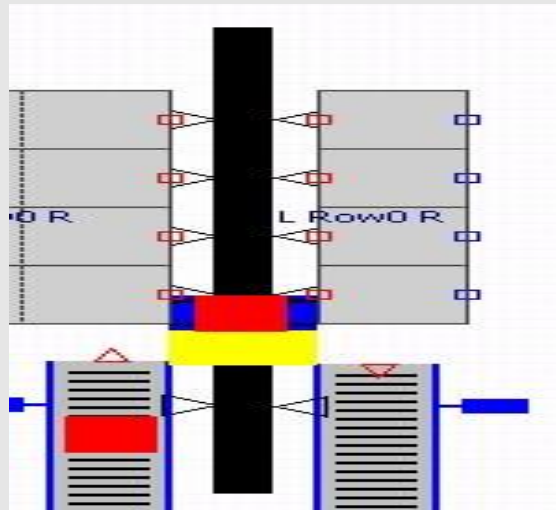
The storage and retrieval machines and racks form the backbone of an automated warehouse system. Automated storage and retrieval machines are responsible for storing and retrieving transport units from and to rack bays. WirthSim supports multi-depth storage bays (n). The automated storage and retrieval machine generally supports two speed settings (loaded and unloaded AS/RS fork).

WirthSim supports the following types:

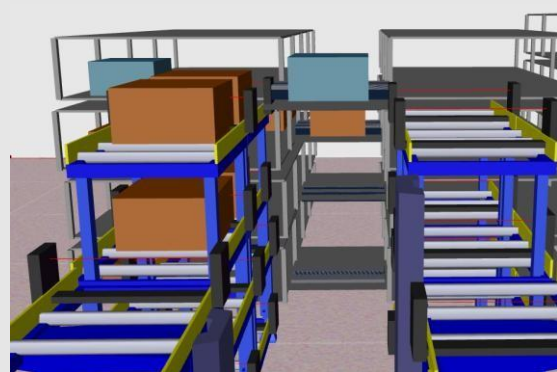
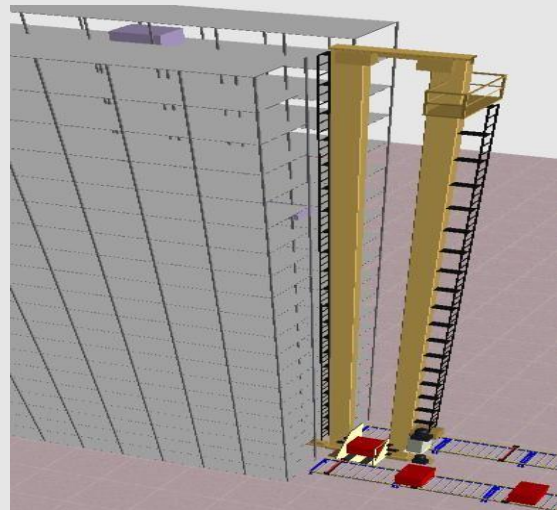
- ▶ Aisle-bound stacker cranes (one crane operates in the aisle)
- ▶ Aisle-bound stacker cranes (two units operate in the aisle)
- ▶ Aisle changers (curved stacker cranes) ▶ Shuttle systems

Rack Servicing Vehicle & Shuttle

2D design view



3D view



The stacker crane and shuttles are used for the storage and retrieval of transport units into and out of racks/storage bins. It supports multiple-depth Storage racks. The element can be configured with two sets of parameters (speeds) for operation in both loaded and unloaded conditions.

The following configurations are possible:

-  One or two forks (configurable)
-  Handling of one or two transport units on an RBG fork (configurable)
-  Shuttle handling

This element is used in conjunction with racks. It enables the simulation of putaway and retrieval operations within automated warehouse systems.

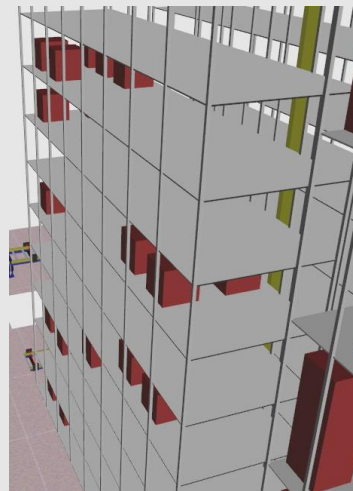
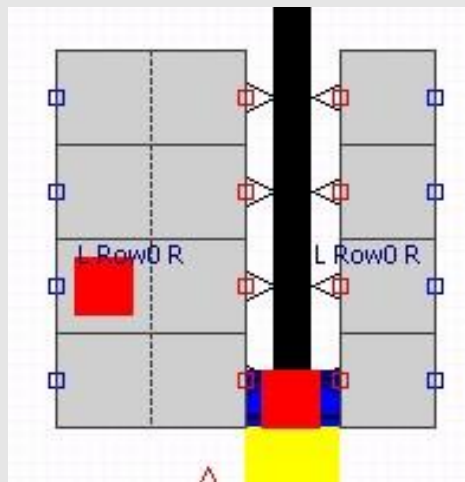
Description			
Element supports the following:			
Variable length, width, height, and angle			
Variable fork length and height			
Unloaded Parameters (Empty Fork)			
Acceleration (m/s^2), deceleration ramp (m/s^2), and speed (m/s) of the X, Y, and Z axes			
Loaded parameters (fork loaded)			
Acceleration (m/s^2), deceleration ramp (m/s^2), and velocity (m/s) of the X, Y, and Z axes			
Stop positions around this element with racks and conveyor elements			
Definition of the number of RBG forks (max. two)			
Definition of the number of transport units per RBG fork (max. two)			
Specification regarding stacker crane type			
Picking mode with 2 TEs per RBG fork			
Setup for Picking Mode with 2 TEs per RBG Fork			
Fork Spacing (mm)			
Timeout for the second TE when there are 2 TEs per RBG fork			
Eco-Sim			
X-axis power consumption (kW)			
X-axis peak power (maintained for a maximum of 1.4 seconds)			
X-axis Braking Power (kW)			
Y-Axis Power Consumption, Acceleration, Upward Direction (kW)			
Y-axis energy peak (during acceleration, remains at this level for a maximum of 1.4 seconds)			
Y-axis power consumption during braking, upward direction (kW)			
Y-axis power consumption at maximum speed, upward direction (kW)			
Y-axis power consumption during acceleration, downward travel (kW)			
Y-axis power consumption during braking, downward direction (kW)			
Y-axis power consumption at maximum speed in the downward direction (kW)			

Z-axis power consumption at max. speed			
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12.2 Symmetrical Frame

2D development view

3D view



The symmetrical rack is used in automated warehouses for storing transport units (pallets, containers, etc.). This rack type supports only symmetrical rack bays. All bays have the same dimensions in height, length, and width. Multiple different defined rack “blocks” can be connected to a rack servicing device or shuttle. This means that not all racks connected to a rack servicing device or shuttle need to have the same dimensions.

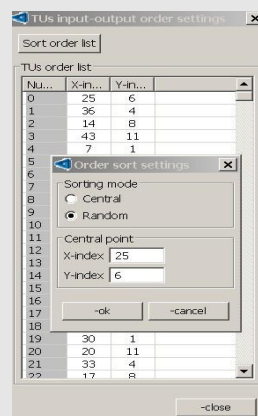
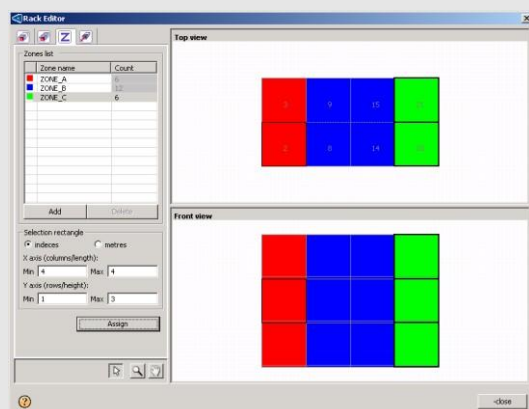
Racks can be assigned n-X, n-Y, and n-Z addresses. A dialog box assists the operator in defining storage zones within the racks. Furthermore, it is possible to define picking and putaway algorithms (transport orders) as well as to create FEM definitions.

The warehouse (racks) forms the backbone of an automated warehouse system. It is not uncommon for warehouses with over 30,000 storage locations to be built.

Description			
Element supports:			
Variable length, width, height, and angle of the frame block			
Variable length, width, and height of the storage compartments			
Variable spacing between storage compartments (crane fork)			
Definition of storage zones			
Definition of the Picking Algorithm for All Destinations			
Definition of rack groups (used for PRA rules)			
3D display settings (complex or simple) => Graphics card			
Static or Dynamic Racks			
Create TEs for stock removal or wait until TEs have been put into storage.			
The "Connection Group ID" is used for shuttle systems or when multiple shuttles are stacked on top of each other. Shuttle rails, conveyors, and racks can be combined into a single group.			

12.2.2 Dialog Boxes

Definition of Zones Within Frames and FEM Simulation



Creating transport orders (source: rack)

Transportaufträge konfigurieren

7:0:0 to 8:0:0 / 22
7:0:0 to 8:0:0 / 200

Transportaufträge | Transporteinheiten | Verteilung

Intervall Startzeit: 7 : 0 : 0
Intervall Stoppzeit: 8 : 0 : 0
Anzahl TEs: 200 Alle Transporte selektieren

Startzeit	Quelle	Ziel	Breite	Länge
07:00:18.000	L10-105-0000-000	L10-106-0001-000	600	600
07:00:35.900	L10-105-0000-000	L10-106-0001-000	600	600
07:00:53.800	L10-105-0000-000	L10-106-0001-000	600	600
07:01:11.700	L10-105-0000-000	L10-106-0001-000	600	600
07:01:29.600	L10-105-0000-000	L10-106-0001-000	600	600
07:01:47.500	L10-105-0000-000	L10-106-0001-000	600	600
07:02:05.400	L10-105-0000-000	L10-106-0001-000	600	600

Ziel der Transporteinheiten (TEs)

Farbe	Typ	Bezeichnung	Nummer
■	Exit	L10-106-0000-000	10-106-0000-000
■	Exit	L10-106-0001-000	10-106-0001-000

Pfad: 1001 from: 10-105-0000-000 to: 10-106-0001-000 distance: 11275.00

Neu Entfernen
Import Export

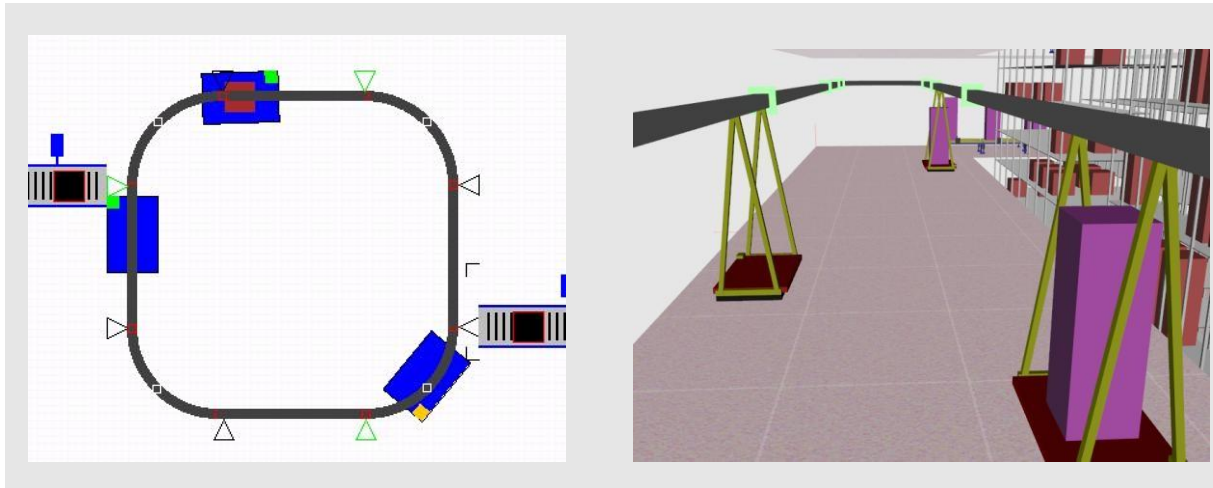
SPEICHERN

Schliessen

13 Monorail (electric overhead conveyor) elements

2D Development View

3D view

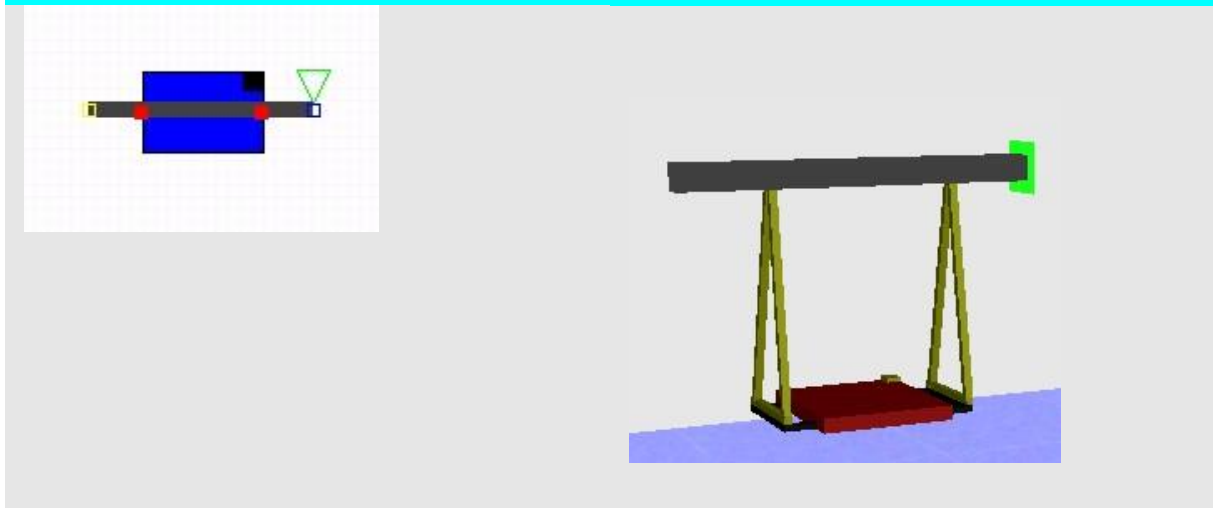


Monorail systems are typically used for transporting heavy goods and/or supplying assembly stations. The speed of the vehicles ranges from 1 to 2 m/s. The track (blocks) can be mounted on the floor or on the ceiling.

13.1 Vehicle 1. TE

2D development view

3D view



Vehicle 1 TE carries a maximum of one transport unit. It is connected to the block element (above) and contains a motor for propulsion as well as a conveyor drive.

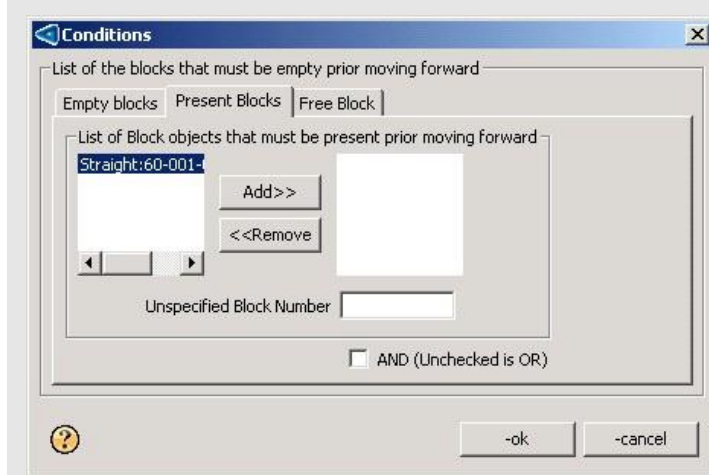
Some vehicle types run on a rail attached to the floor. The monorail system supports both the so-called “block” mode and distance-oriented operation (higher speed => greater vehicle spacing).

Monorail vehicles transport heavy cargo over long distances. They are typically used in large-scale warehouse systems.

Description			
Element supports:			
Variable length, width, height, and angle			
Variable times for loading and unloading transport units			
Definition of the maximum speed when the vehicle is loaded or unloaded			
Variable time when delivery is not possible (=> one lap)			
Definition of dependencies, e.g., next block free, branch...			
Setup of "Block" or "Distance Handling"			
Eco-Sim			
Energy Consumption of the System (kW)			
Braking Energy Consumption (kW)			
Energy consumption at v-max. (kW)			

13.1.2 " " dialog boxes

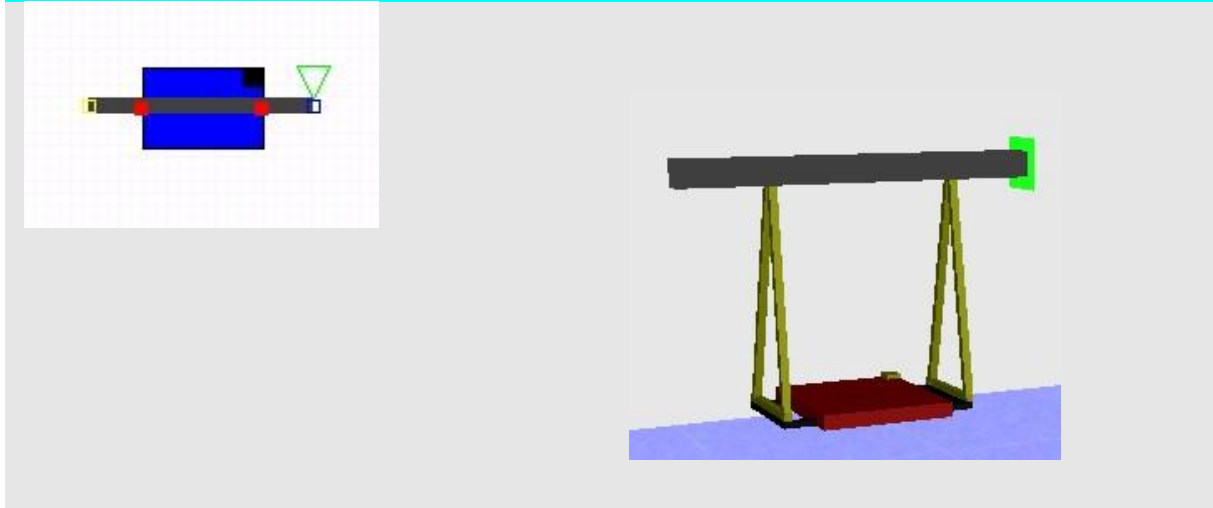
Define Dependencies



13.2 Vehicle 2. TE

2D Development View

3D view

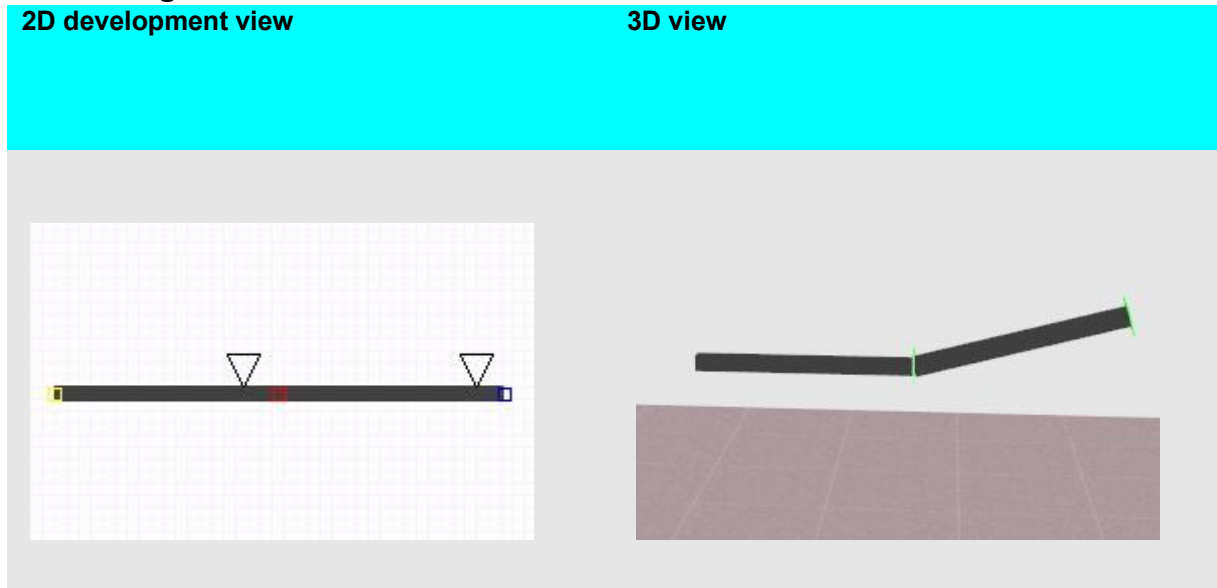


The Vehicle 2 TE carries a maximum of two transport units. All other parameters are the same as for the vehicle with one transport unit.

13.3 Straight Block

2D development view

3D view



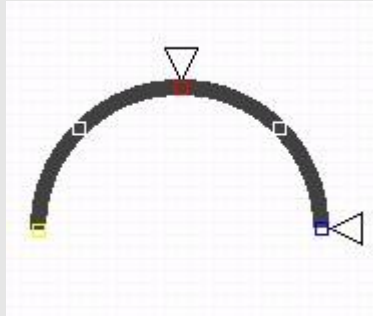
The straight block serves as the base from which the vehicles are suspended. These vehicles follow the tracks. Each stop position of a block can be connected to a conveyor element.

Description			
Element supports:			
Variable length, height, and angle			
Acceleration (m/s^2), deceleration ramp (m/s^2), and velocity (m/s) of the element			
Definition of the minimum distance between the vehicles			

13.4 Curve Block

2D development view

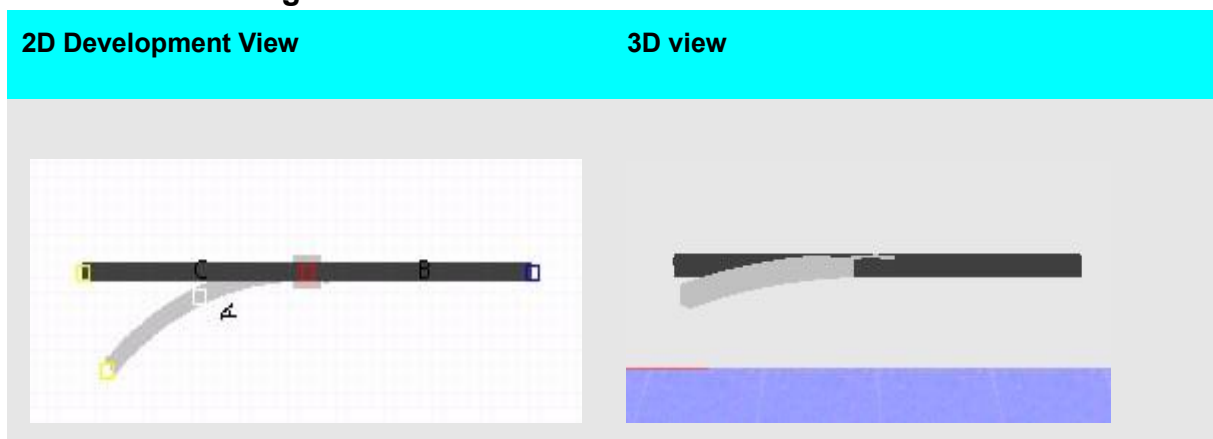
3D view



The curve block forms the base from which the vehicles are suspended. They follow the tracks. Each block of this type has only one stop position.

Description			
Element supported in:			
Variable length, height, and angle			
Acceleration (m/s^2), deceleration ramp (m/s^2), and velocity (m/s) of the element			
Definition of the minimum distance between vehicles			
Definition of the curve radius			

13.5 Block Merge



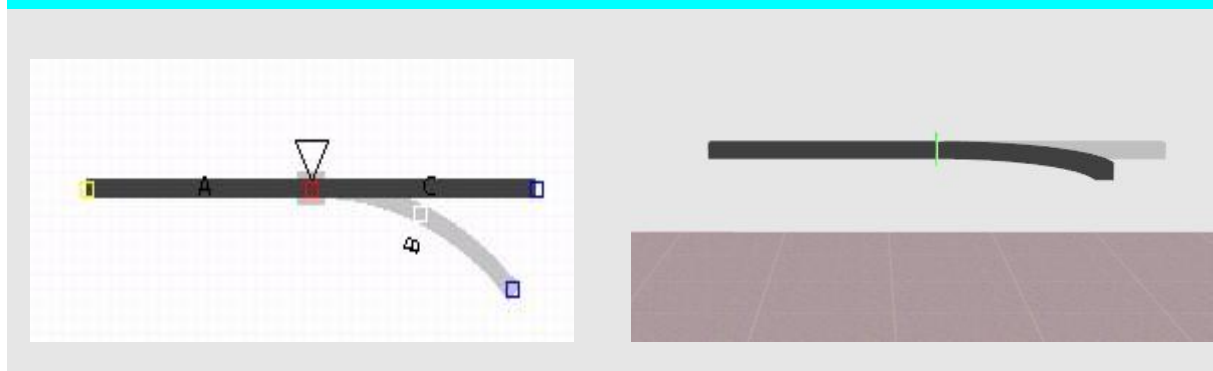
The Merge Block element is the base to which the vehicles are attached. It is used to merge two paths. Each block of this type has only one stop position.

Description			
Element supported in:			
Variable length and height			
Acceleration (m/s^2), deceleration ramp (m/s^2), and velocity (m/s) of the element (three different setups)			
Definition of the minimum distance between vehicles			
Definition of the curve radius			

13.6 Branch Block

2D Development View

3D view

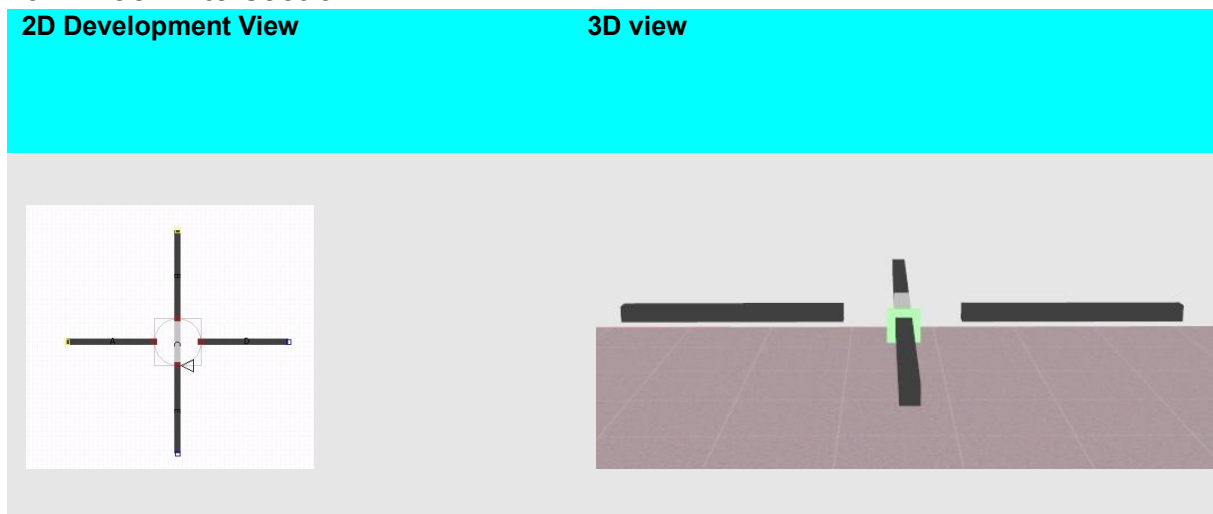


The Branch Block element is the base to which the vehicles are attached. It is used to branch off into two paths. Each block of this type has only one stop position.

Description			
Element supported in:			
Variable length and height			

Acceleration (m/s^2), deceleration ramp (m/s^2), and speed (m/s) of the element (three different setups)			
Definition of the minimum distance between the vehicles			
Definition of the curve radius			

13.7 Block Intersection



The Block Junction is the base to which the vehicles are attached. It serves as a junction. Each block of this type has only one stop position.

Description			
Element supported in:			
Variable length and height			
Acceleration (m/s^2), deceleration ramp (m/s^2), and velocity (m/s) of the element (three different setups)			
Definition of the minimum distance between the vehicles			

13.8 Lift

2D development view

3D view



The Monorail Lift Element connects different floors and bridges level differences within floors. It can be used as follows:

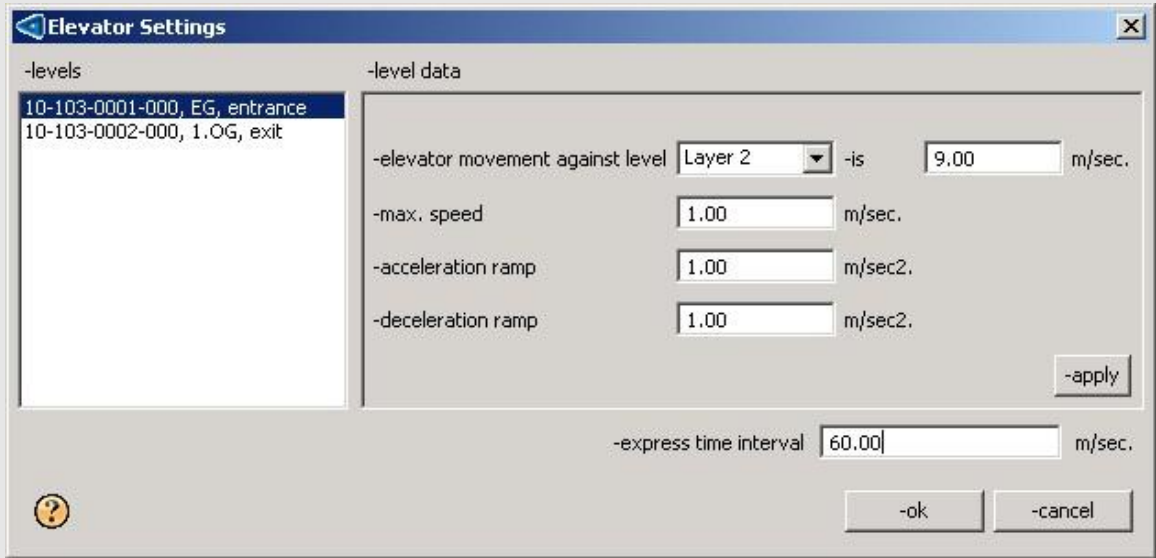
- ▶ Input only (transport units are loaded)
- ▶ In and Out (cargo is loaded into and unloaded from the lift) ▶ Out Only (cargo can only exit the lift).

Description			
Element is supported in:			
Variable width and angle			
Variable Element Height (Conveyor)			
Acceleration (m/s^2), deceleration ramp (m/s^2), and speed (m/s) of the conveyor element			
Lift travel times to and from each floor or landing			
Rest position of the lift (floor / landing)			
Lifter operating modes: pick up only, pick up and set down, and set down only			

Priority Transports (Based on Time)			
Delay time			

13.8.2 Dialog Boxes

Definition of travel paths between floors or landings



Elevator Settings

-levels

- 10-103-0001-000, EG, entrance
- 10-103-0002-000, 1.OG, exit

-level data

-elevator movement against level Layer 2 -is 9.00 m/sec.

-max. speed 1.00 m/sec.

-acceleration ramp 1.00 m/sec2.

-deceleration ramp 1.00 m/sec2.

-apply

-express time interval 60.00 m/sec.

-ok -cancel